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Investigation on the effect of container configurations and forecastle fairings on wind resistance and aerodynamic performance of large container ships

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ABSTRACT

The wind resistance of a model of a large container ship was investigated experimentally and numerically under various conditions to investigate the aerodynamic performance and identify approaches for reducing wind resistance. The container ship model with and without six types of forecastle fairing was tested in a wind tunnel under conditions of full load and various uneven loads in different wind directions. The model test found that a specially designed forecastle fairing and certain container configurations are effective in reducing wind resistance. Then, a numerical simulation was performed to investigate the characteristics of the flow field around the test model. The comparisons and analyzes of the pressure distribution, streamlines, and vortex structure illuminate the mechanism of load reduction caused by the forecastle fairing and container configurations that are useful for reducing wind resistance. The optimization of a container ship for wind resistance is dominated by the effect of the container configuration; the contributions of forecastle fairing are secondary. When the overall effects of forecastle fairing and container configuration are considered, the streamlined load is the variant most optimized for wind resistance.

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KEYWORDS

Container ship; forecastle fairing; container configuration; vortex interaction; load reduction

1. Introduction

With the implementation of the Energy Efficiency Design Index (EEDI), the requirements for large container ships to save energy and reduce emissions are increasing yearly. Owing to the large windward area of a container ship, wind resistance can account for 2–10% of the ship's total resistance (Majidian & Azarsina, 2018). The wind resistance of a large container ship accounts for a greater proportion of the total resistance and influences the effectiveness of the ship's propellers, maneuverability in heavy winds, and related characteristics (Andersen, 2013). Computational fluid dynamics (CFD) simulations conducted for various hull geometries (Islam & Soares, 2019; Janssen et al., 2017; Rastgou & Saedodin, 2013) have shown that the difference in wind resistance between different container ships reaches 5.9–37.9%. Nguyen and Ikeda (2017a) studied the wind resistance of container blocks with gaps between them. Their numerical examination of the effects of a partial gap between blocks found that a 50% partial gap cover reduces the wind resistance in 30° winds by 19%. Hassan et al. (2012) studied the power optimization of a container ship using CFD numerical simulation, finding that an optimal container stack arrangement can reduce wind resistance by 30%. As

the large frontal projected area surfaces above water surface, research on reducing wind drag acting on hull of the ship is very important (Van He et al., 2020).

A wind tunnel was utilized to estimate the wind resistance of marine structures in Isherwood (1972). Many ship geometry optimization methods have been presented and analyzed, using experiments employing towing tanks or wind tunnels to obtain the ship resistance values (Liu et al., 2016; Niklas & Pruszko, 2019; Roberston et al., 2011; Wnek & Soares, 2015). Andersen (2013) studied the influence of the container configurations of a 9000 TEU container ship using a wind tunnel and found that the ship's wind resistance was determined mainly by the container configurations, which suggests an idea for studying the wind resistance of a container ship. In a theoretical and experimental investigation of the wind resistance acting on a 20,000 TEU container ship (Watanabe et al., 2016). Although research on the wind resistance of large container ships by wind tunnel is important for improving their aerodynamic performance, Their discussion of the reasons for the wind drag is not supported by computational dynamics methods.

Ricci et al. (2020) studied the wind resistance of a container ship with various configurations created by

changing the positioning of containers on the deck and presented a case study in which the wind resistance of a large cruise ship moored at the quay of the Rotterdam Cruise Terminal was determined by 3D steady Reynolds-averaged Navier–Stokes (RANS) simulations, showing that the presence of high-rise buildings can produce locally amplified surface pressures. The effects of various container ship shapes on wind drag have been assessed using computational fluid methods (Acanfora et al., 2017; Stern, 2006). Mou et al. (2017) ascertain wind pressure distributions on and around various squared-shaped tall buildings by the application of Computational Fluid Dynamics techniques and wind pressure distributing on buildings was predicted under two scenarios. The flutter behavior of a typical wing is investigated by presented Deferential Quadrature Method (DQM). Quasi steady and unsteady aerodynamics are considered to estimate the instability speed of the structure (Ghalandari et al., 2019). Simulations of the flow around a generic container freight wagon model, lorry model, and models of other structures (Hemida & Baker, 2010; Kääriä et al., 2013; Lo & Kontis, 2017; Östh & Krajnović, 2014; Viola et al., 2015) were performed to predict the aerodynamic forces. For example, A CFD-based model has been developed for predicting the aerodynamic forces on the rig and sails of the U.S. Brig Niagara and Wind tunnel tests and full-scale experiments were performed to validate the model (Lasher & Flaherty, 2009). Li et al. (2021) had a numerical investigation on the aerodynamic resistances of double-unit trains with different gap lengths. They can accurately obtained the wind resistance values of ships or other structures.

CFD results revealed that the airflow through a container ship is very complex and includes many types of vortices. Only a limited number of studies have analyzed the effects of forecastle fairing and container configurations. Yokoyama et al. (2013) discussed the vortical structures in the forward flight of a wing-flapping butterfly and studied the vortex characteristics. Nguyen and Kinugawa (2017b) investigated practical gap covers to reduce the wind resistance acting on deck containers of a ship; they confirmed numerically that the wind resistance due to the airflow in gaps between containers is caused by the difference in pressure just near the gap entrance. It seems that airflow through the gaps between container blocks plays an important role in the hydrodynamic characteristics of wind resistance. Van He et al. (2016, 2019) presented research on the interaction between the hull and accommodation on aerodynamic performance and wind drag acting on a wood chip carrier hull. Van He et al. (2020) investigated the interaction between the hull and accommodation in the effect on wind drag for a full-scale container ship of 1200 twenty-feet equivalent

units (TEUs). They found that a drastic reduction in the combined effect of the hull and accommodation on the wind drag acting on the ship could be achieved. However, The results were inconclusive, and the vortical structures associated with the container ship require further investigation.

The container ships with various forecastle fairings and container configurations were investigated experimentally and numerically. The influence of wind directions, pressure, streamlines and vortex interactions are concerned. The wind tunnel setup, ship geometry, container configurations and numerical method used in the study are introduced in Section 2. In Section 3, the experimental results with regard to wind resistance are presented and discussed. The verification and validation are introduced. In Section 4, we present the results of our numerical investigation based on large-eddy simulation (LES) to characterize the flow fields around large container ships. The cases of the forecastle fairings and container configurations are discussed in three part. The typical pressure distributions, streamlines, and vortex interactions are analyzed and discussed in detail.

2. Materials and methods

2.1. Wind tunnel specifications, model geometry, and load configurations

The effect of the deck container configuration and of fairing on the wind resistance of the 1900 TEUs container ship was experimental investigated in the wind tunnel of Wind Tunnel and Water Flume (WTWF) laboratory at the Harbin Institute of Technology. The type of wind tunnel used in our test is closed return tunnel, which is closed and re-circulates the air through the test section. The closed single-return wind tunnel used has two experimental sections (The big testing section, which is 50.0 m long, 6.0 m wide and 3.6 m high, and The small testing section, which is 25.0 m long, 4.0 m wide and 3.0 m high) and a water flume (The tank, which is 50.0 m long, 5.0 m wide and 4.5 m high). A turntable system, the diameter of which is 2.5 m, is installed in the test section to allow the direction of the test model to be changed. The wind velocity can be set to values between 3.0 m/s and 50.0 m/s, with velocity heterogeneity and turbulence intensity values of less than 1% and 0.46% respectively. In this paper, the Wind resistance test was carried out in the small testing section. The test model, made of fiberglass, had a scale ratio of 1:60; its length, width, and freeboard were 2.41 m, 0.39 m, and 0.13 m, respectively. The projected areas in the longitudinal, transverse, and vertical directions were 0.15 m², 0.61 m², and 0.92 m², respectively, in the full-load condition. The wind velocities adopted for

Table 1. Parameters of container ship model under full load.

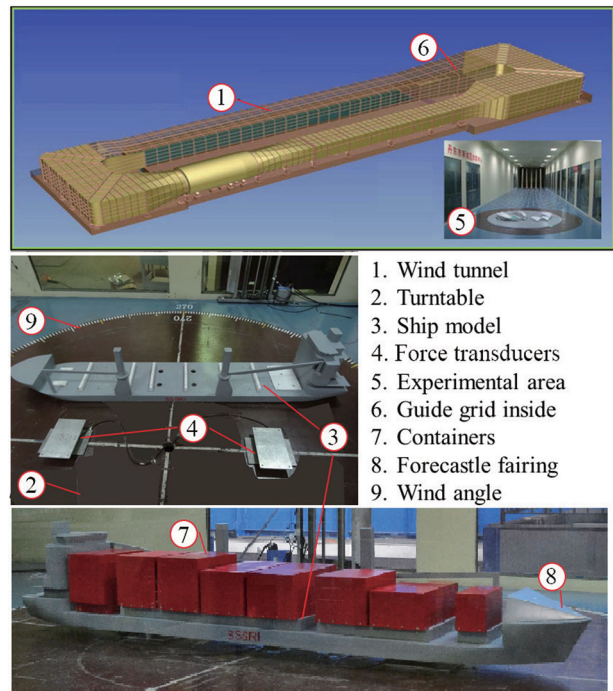
Parameter	Description	Value (full-scale ship)	Value (model)	Unit
L	Length	144.80	2.41	m
B	Breadth	23.40	0.39	m
T	Freeboard	8.0	0.13	m
A_x	Frontal area	–	0.15	m ²
A_y	Side windward area	–	0.61	m ²
A_z	Projected area above waterline	–	0.92	m ²
Re	Reynolds number	–	4.22×10^6	–

the model testing were 25.0 m/s and 15.0 m/s, and the Reynolds numbers were 4.22×10^6 and 3.0×10^6 . These parameters are listed in Table 1.

The purpose of this research was to investigate the effect of forecastle fairing and container configuration on a container ship's wind resistance and aerodynamic performance, and it is incontrovertible that the effect of vortices cannot be ignored. To mitigate the vortex generated by flutter of the model in the test, the model was supported by two six-axis force/torque transducers (ATI Delta; Measurement accuracy: ± 0.125 N) located 1/4 ship length from the bow and from the stern. The sampling rate of the Transducers using in our test is 1600 Hz. the area of orthogonal projection above waterline of the model is 0.144 m^2 , and the area of lateral projection above waterline is 0.603 m^2 . The cross-sectional area of the small testing section of the wind tunnel is 12 m^2 . The blockage ratio of the wind tunnel for our model is 0.12–5.025%, which meet the test requirements of 5–7% in the low speed industrial Aerodynamics (Wind Engineering) Wind tunnel (Fan, 2002). The transducers were calibrated before the test, and the resultant wind resistance is the sum of the data obtained by the two transducers at each reading. The experimental setup is shown in Figure 1.

Containers were placed in stacks on the deck. The maximum number of containers was 65 in the longitudinal plane and 9 in the cross section. The six forecastle fairings were divided into two design groups of three models each and are shown in Figure 2. The size of a single container is the standard 20GP (the external dimension $20.0 \text{ ft} \times 8.0 \text{ ft} \times 8.5 \text{ ft}$).

Cases A–G were for the ship model with 522 containers (a full load); in Case A there was no forecastle fairing, and in each of Cases B–G a fairing was mounted at the bow. For Cases B–D, the length and height of the forecastle fairings were 257.2 mm and 81.7 mm, respectively, and for Cases E–G, 307.2 mm and 97.6 mm, respectively. As shown in Figure 2, the difference between the two groups is that the forecastle fairings in the second group are obtained by extending the forecastle fairings in the first group correspondingly; the profiles of the corresponding

**Figure 1.** Wind tunnel and ship model.

forecastle fairings are the same. The extended forecastle fairings are used to cover the gaps between the containers.

Because of the offloading of containers in different ports of call in a row, ship owners' requirements and so on, container ships are not in a fully loaded condition. It is therefore important to understand the aerodynamic performance of an unevenly loaded container ship and investigate approaches for reducing its wind resistance, especially because the wind resistance of an unevenly loaded container ship is normally greater than that of a fully loaded container ship (Andersen, 2013). Thus, in addition to the cases for the fully loaded container ship with and without forecastle fairings, the test included cases for an unevenly loaded model with no forecastle fairing under different conditions and with different wind directions. The assumption for constant draft and freeboard was included for loading configurations. The full list of test cases is given in Table 2.

To investigate the wind resistance and aerodynamic performance of the container ship in conditions of uneven load, 178 containers were removed, as several typical configurations can be constructed using just the remaining 344 containers (65.91% of a full load) by using an elaborate design. Several typical configurations having the same maximum number of containers in different uneven configurations were compared, including four cases with various randomly distributed loads (Cases H–K) and one case with a streamlined load (Case L). The specifications for these cases are illustrated in Figure 3.

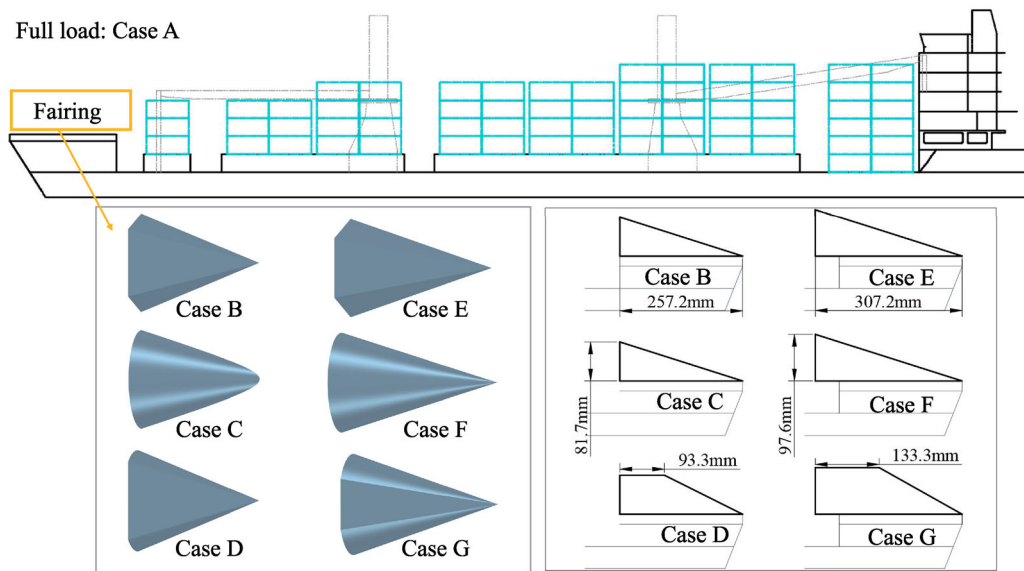


Figure 2. Ship model for Case A and forecastle fairings for Cases B–G. Left, fairing types; right, side view of geometry.

Table 2. Parameters of test cases.

Case	Fairing	Configuration	Number of containers
Case A	Without fairing	Full load	522
Case B	Fairing 1	Full load	522
Case C	Fairing 2	Full load	522
Case D	Fairing 3	Full load	522
Case E	Fairing 4	Full load	522
Case F	Fairing 5	Full load	522
Case G	Fairing 6	Full load	522
Case H	Without fairing	Randomly distributed load	344
Case I	Without fairing	Randomly distributed load	344
Case J	Without fairing	Randomly distributed load	344
Case K	Without fairing	Randomly distributed load	344
Case L	Without fairing	Streamlined load	344

Some configurations of randomly distributed loads are very uneven, with one or more completely empty bays adjacent to each other or distributed over the deck surface. Gaps in the container blocks were formed between the bays. Among the four cases with irregularly distributed containers, Case H had a maximum stack height of five containers; in Case I, the containers were used to fully load each bay from the stern; in Case J, the containers were used to fully load each bay from the bow and stern; and in Case K, the containers were used to fully load each bay from the bow. In Case L, the streamlined load configuration, the height of the stacks increases from the bow to the stern, and the last two bays are fully loaded. The difference between these cases is the height of the stacks and the locations of the container blocks, and an understanding of the relationship between the container configuration and the ship's aerodynamic performance will be useful for reducing wind resistance. Note that the focus of the present study is the aerodynamic performance of the container ship; the effect

of the container distribution on the hull structure and navigational attitude is ignored.

2.2. Experimental setup and conventions

The test model is heading in the positive X direction in the Cartesian coordinate system, starboard is the positive Y direction, and the draft is in the positive Z direction. When the wind is in the negative X direction, the wind direction is 0° , and when the wind is in the positive X direction, the wind direction is 180° . The wind direction may range from 0° to 180° at 15° intervals and is adjusted using the turntable system. A dual-transducers mechanism was used during measurements to prevent the load measured by a single transducer from exceeding the maximum measurable values. The dual-transducers setup was fixed at two ends around the center of a rotating platform with transducers 1 and 2 being placed at distances l_1 and l_2 , from the center, respectively. The ship model was placed horizontally inside the wind tunnel atop the dual-transducers setup. The sum of the measured wind resistances obtained from both transducers was recorded as the measured result during each trial. The definition of the wind direction with respect to the ship model and its coordinate system is shown in Figure 4.

2.3. Numerical method

In order to analyze the effect of forecastle fairing and container configuration on the wind resistance, a numerical simulation was conducted using the Star-CCM+ (Version 15.02) software to investigate the details of the viscous flow field around the model. The LES and wall-adapting

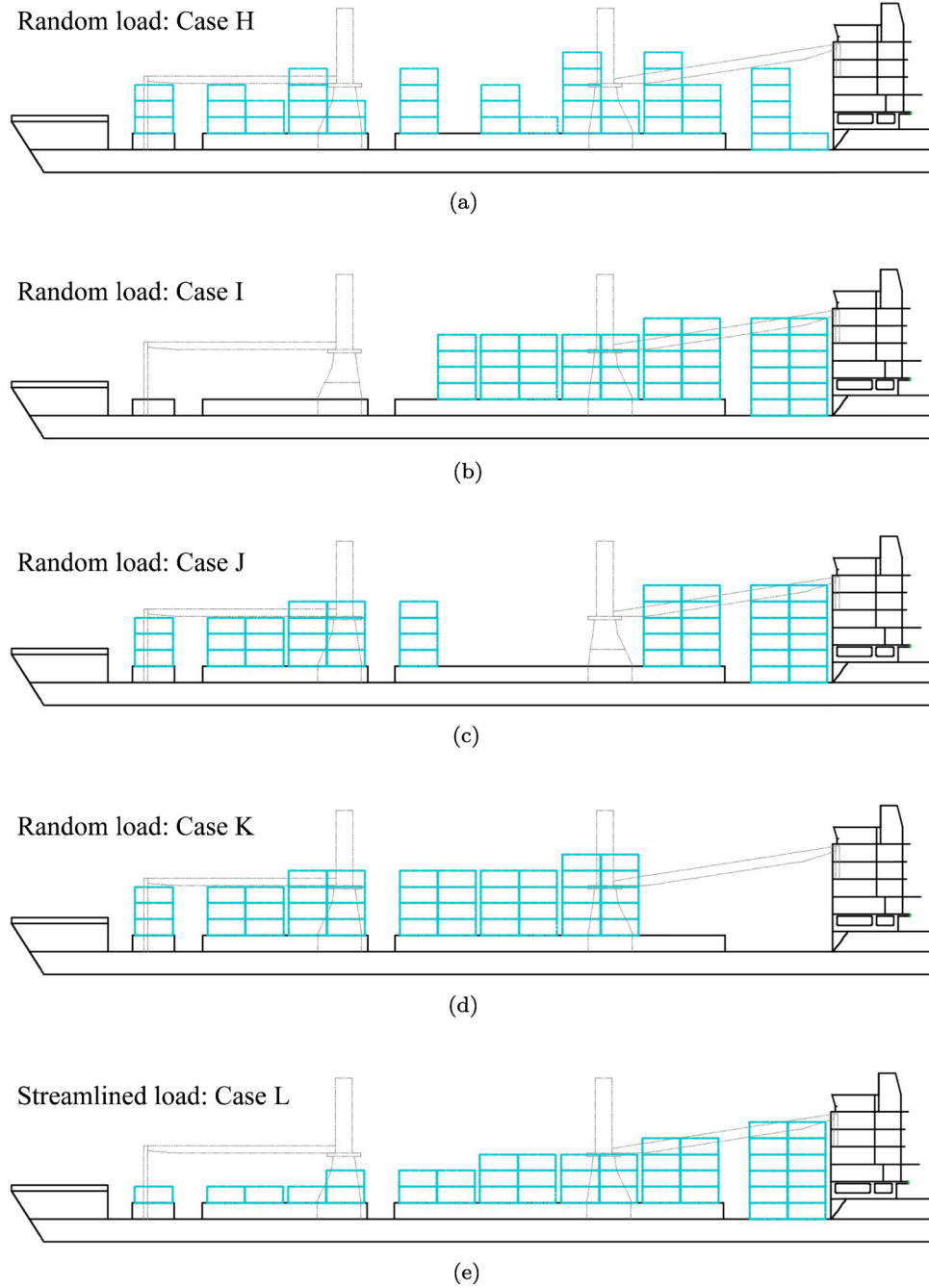


Figure 3. Configurations of ship model in the cases with uneven loads. (a) Case H (with 344 containers, randomly distributed load). (b) Case I (with 344 containers, randomly distributed load). (c) Case J (with 344 containers, randomly distributed load). (d) Case K (with 344 containers, randomly distributed load) and (e) Case L (with 344 containers, streamlined load).

local eddy-viscosity (WALE) model were adopted for the simulation. The large-scale vortices can be solved directly by the Navier–Stokes equation, whereas the small-scale vortices need to be solved by subgrid scale models. In LES, the filtered mass and momentum conservation equations can be written as

$$\frac{\partial \bar{U}_i}{\partial x_i} = 0 \quad (1)$$

$$\frac{\partial \bar{U}_i}{\partial t} + \frac{\partial \bar{U}_i \bar{U}_j}{\partial x_j} = -\frac{1}{\rho} \frac{\partial \bar{P}}{\partial x_i} + \nu \frac{\partial}{\partial x_j} \left(\frac{\partial \bar{U}_i}{\partial x_j} + \frac{\partial \bar{U}_j}{\partial x_i} \right) \quad (2)$$

The non-linear term $\bar{U}_i \bar{U}_j$ is defined as

$$\begin{aligned} \bar{U}_i \bar{U}_j &= \overline{(\bar{U}_i + \bar{U}_i') (\bar{U}_j + \bar{U}_j')} \\ &= \bar{\bar{U}_i \bar{U}_j} + \bar{\bar{U}_i \bar{U}_j'} + \bar{\bar{U}_i' \bar{U}_j} + \bar{\bar{U}_i' \bar{U}_j'} \end{aligned} \quad (3)$$

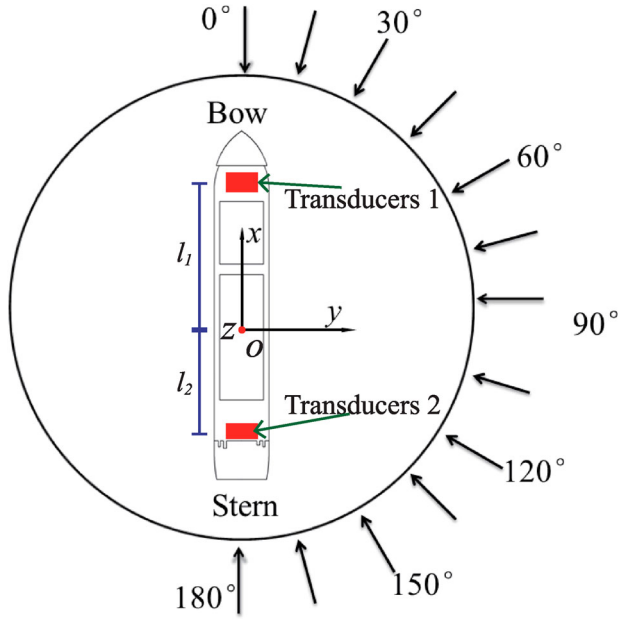


Figure 4. Definitions of the coordinate system and wind direction.

The subgrid scale stress τ_{ij} is defined as

$$\begin{aligned}\tau_{ij} &= L_{ij} + C_{ij} + R_{ij} \\ &= \overline{U_i U_j} - \overline{U_i} \overline{U_j} + \overline{U_i U_j'} + \overline{U_j U_i'} \\ &\quad + \overline{U_i' U_j'} - \overline{U_i} \overline{U_j} - \overline{U_i} \overline{U_j} \end{aligned} \quad (4)$$

where the stress tensor $L_{ij} = \overline{U_i U_j} - \overline{U_i} \overline{U_j}$ is the integration in the large scale and small scale. The Reynolds subgrid tensor $C_{ij} = \overline{U_i U_j'} + \overline{U_j U_i'}$ is the integration between subgrid scales. The Leonard tensor $R_{ij} = \overline{U_i' U_j'}$ reflects the integration between subgrid scales. According to the definition of the subgrid scale stress, the Navier-Stokes equation in LES is defined as:

$$\begin{aligned}\frac{\partial \overline{U_i}}{\partial t} + \frac{\partial \overline{U_i U_j}}{\partial x_j} \\ = -\frac{1}{\rho} \frac{\partial \bar{P}}{\partial x_i} + \nu \frac{\partial}{\partial x_j} \left(\frac{\partial \overline{U_i}}{\partial x_j} + \frac{\partial \overline{U_j}}{\partial x_i} \right) - \frac{\tau_{ij}}{\partial x_j} \end{aligned} \quad (5)$$

The subgrid scale stress can be written as follows using the Boussinesq hypothesis:

$$\tau_{ij} - \frac{1}{3} \tau_{kk} \delta_{ij} = -\nu_t \left(\frac{\partial \overline{U_i}}{\partial x_j} + \frac{\partial \overline{U_j}}{\partial x_i} \right) \quad (6)$$

where $\mu_t = \rho \Delta^2 S_w$ is the subscale turbulence viscosity, in which Δ is the length scale or grid filter width and S_w is the deformation parameter, defined as

$$\Delta = \min(\kappa d, C_w V_c^{1/3}) \quad (7)$$

$$S_w = \frac{\overline{S_d} : \overline{S_d}^{3/2}}{\overline{S_d} : \overline{S_d}^{5/4} + \overline{S} : \overline{S}^{5/2}} \quad (8)$$

where the von Kármán constant is $\kappa = 0.41$, d is the distance to the closest wall, V_c is the cell volume, and the coefficient C_w is 0.554. The strain rate tensor of the filtered flow $\overline{S}$ and the tensor $\overline{S_d}$ are respectively defined as

$$\overline{S} = \frac{1}{2} \left(\nabla \overline{u} + \nabla \overline{u}^T \right) \quad (9)$$

$$\overline{S_d} = \frac{1}{2} \left(\nabla \overline{u} \nabla \overline{u} + \left(\nabla \overline{u} + \nabla \overline{u}^T \right) \right) - \frac{1}{3} \text{tr}(\nabla \overline{u} \cdot \nabla \overline{u}) I \quad (10)$$

where I is the identity tensor.

2.4. Computational domain and boundary conditions

To resolve the crucial turbulent structures near the surfaces of a container ship, LES needs to have an excessively high mesh resolution in the wall boundary layer and the flow direction. Figure 5 shows a sketch of the computational domain, the dimensions of which are similar to the test area of the wind tunnel in the experiment: Length $8.0L$, Width $3.0L$, Height $1.5L$. The bow is positioned $2.5L$ from the velocity inlet.

Figure 6 shows the details of the mesh from the three perspectives and the boundary layer near the container ship. The computational domain uses an unstructured mesh for the mesh trimmer. The regions near the container ship and at the rear of the ship have a locally refined mesh to achieve the high resolution needed.

The three types of meshes shown in Table 3 were generated, and the results obtained with them were used to validate the numerical simulation. For the computational domain, the meshes generated were of three types: coarse, medium, and fine. The mesh sizes dx were 0.008 m, 0.004 m and 0.002 m respectively. The refinement rate between different meshes ($dx_{\text{medium}}/dx_{\text{coarse}}$) is 0.5; for the three mesh refinement areas of coarse, medium, fine and very fine in the computational domain, the refinement rate is $0.2B$, $0.1B$, $0.05B$ and $0.01B$. The time steps were 0.00032, 0.00016 and 0.00008 respectively, while the Courant–Friedrichs–Lewy (CFL) number was 1. The surface remesher and prism layer mesher models were used to provide good quality meshes in Star-CCM+. The non-dimensional distance y^+ is close to 1. The implicit unsteady algorithm was adopted for the velocity–pressure coupling, and the second-order scheme with 10 inner iterations per time step was used for the temporal discretization. A similar numerical scheme had

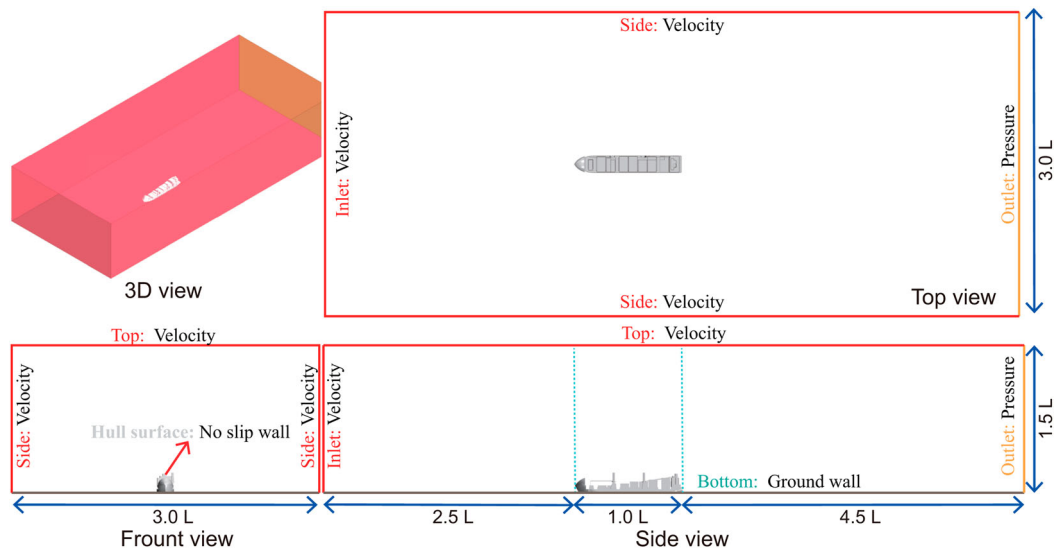


Figure 5. Computational domain.

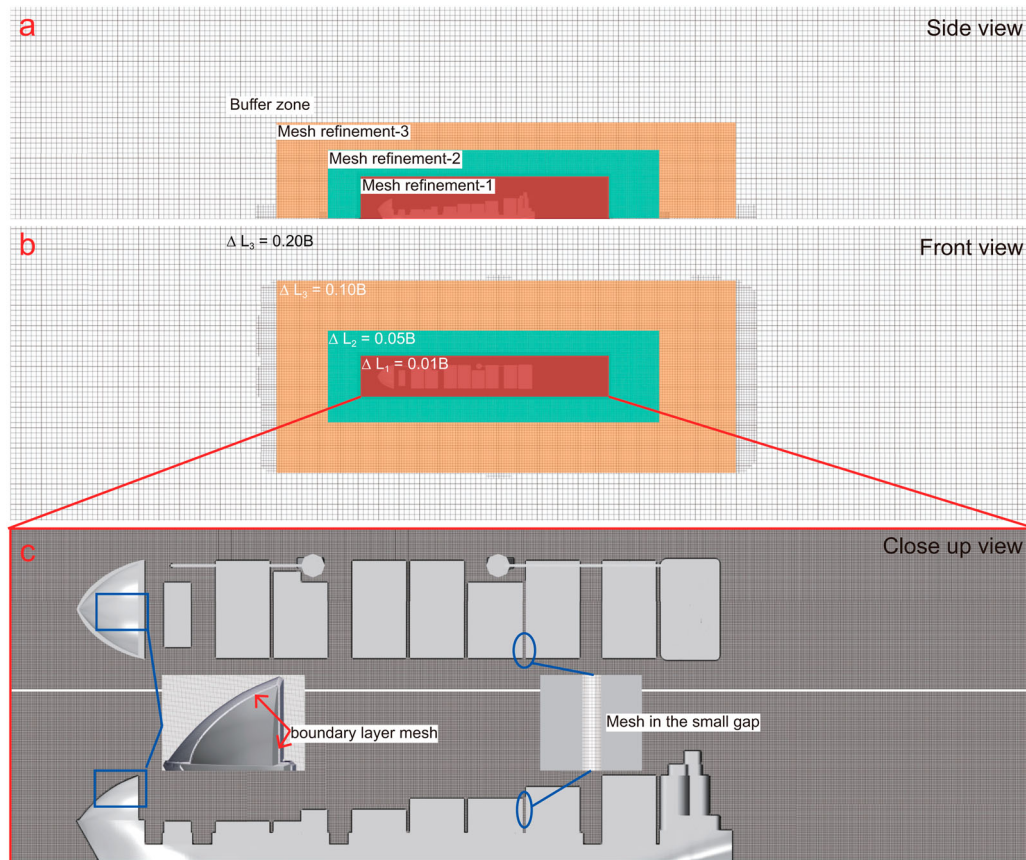


Figure 6. Mesh topology: (a) side view of mesh for the entire computational domain, (b) top view of mesh for the entire computational domain, (c) close up view of mesh positioned on the ship surface.

been used to investigate the flow field of a generic container freight wagon (Östh & Krajnović, 2014). Under-relaxation factors of 0.7 and 0.2 were used for the velocity and pressure, respectively. The bounded central differencing scheme was used for the discretization of convection terms, which is a hybrid scheme blending

second-order upwind and central differencing schemes for LES in Star-CCM+. For this study, the upwind blending factor was set to 0.1 to improve the numerical accuracy. Each simulation case was calculated using a two-way 56-core 2.5 GHz Intel(R) Xeon(R) Platinum 8180 processor.

Table 3. Numbers of meshes and time resolutions.

Mesh type	Number of meshes	Mesh size $dx(m)$	Time step delta $t(s)$	Numerical results (C_{fx})
Coarse	19,434,322	0.008	0.00032	0.729
Medium	41,179,074	0.004	0.00016	0.735
Fine	69,040,361	0.002	0.00008	0.737

Table 4. Verification of numerical results.

Numerical results	s_1	s_2	s_3	ε_{12}	ε_{23}	R_k
C_{fx}	0.729	0.735	0.737	0.006	0.002	0.333

2.5. Verification and validation

Verification and validation were used to estimate the numerical errors for the fully loaded container ship model with no forecastle fairing (Case A). For estimating the uncertainty in the numerical solution, the numerical results with coarse, medium, and fine grids are denoted by s_1 , s_2 and s_3 , respectively, and ε_{12} and ε_{23} denote the difference between the results using the medium mesh and coarse mesh and that between the fine mesh and medium mesh, respectively:

$$\varepsilon_{12} = s_2 - s_1 \quad (11)$$

$$\varepsilon_{23} = s_3 - s_2 \quad (12)$$

The convergence rate R_k is calculated as:

$$R_k = \frac{\varepsilon_{23}}{\varepsilon_{12}} \quad (13)$$

The convergence rate distinguishes three types of numerical solutions as follows: $0 < R_k < 1$: indicates that the solution converges monotonically, and $R_k \leq 0$: indicates that the solution is of oscillatory convergence $R_k \geq 1$: indicates that the solution is divergent. The values of the parameters are shown in Table 4. They show that the numerical results are monotonically convergent with the size of the meshes, which indicates that the fine mesh used in this study is effective for simulating the flow field around the container ship.

The progression of the wind resistance through time and the wall y^+ for the fully loaded container ship model with no forecastle fairing (Case A) as obtained using various meshes are shown in Figure 7. It can be seen in the figure that the numerical results obtained using different meshes are in good agreement with each other, indicating that the numerical uncertainties in these cases are of the same order.

The results of the numerical simulation were additionally compared with the experimental data. Table 5 presents the time-averaged results of C_{fx} obtained using the three meshes along with the test data. The wind velocity was 25.0 m/s, and the wind direction was 0° . It can be

seen in the table that the discrepancy between the corresponding results decreases with the increased number of the meshes, and the difference between the numerical result obtained using the fine mesh and the experimental result is approximately 2.8%, demonstrating satisfactory agreement.

The effect of the randomly distributed load container ship without forecastle fairing on aerodynamic performances have been investigated by the CFD. Comparing the numerical flow field of the literature (Van He et al., 2020), it can be seen in Figure 8 that our numerical method based on LES can capture more detailed flow field, and the changes, as shown in Figure 9, of the pressure and positive(negative) vortex around container ship are more clearly explained. In all, the numerical results for fines meshes can better capture the flow fields.

3. Experimental results and discussion

3.1. The effect of forecastle fairing in the full load condition

The coefficients of wind resistance are defined as

$$C_{fx} = \frac{F_x}{0.5\rho U^2 A_x} \quad (14)$$

$$C_{fy} = \frac{F_y}{0.5\rho U^2 A_y} \quad (15)$$

where

$$F_x = F_{x1} + F_{x2} \quad (16)$$

$$F_y = F_{y1} + F_{y2} \quad (17)$$

F_x and F_y are the resultant forces in the X and Z directions. F_{x1} , F_{x2} and F_{y1} , F_{y2} are the component forces obtained by the transducers 1 and 2 in the X and Z directions, respectively. ρ is the air density, U is the speed of wind, and A_x and A_y are the projected areas of the hull model in Y-Z and X-Y planes, respectively. A wind speed 25 m/s, with a Reynolds number of 4.22×10^6 , was adopted for the experiment (Case A-G) under different wind directions. Each case was tested up to three times, and the test data were averaged to minimize error caused by the transducers. A comparison of the longitudinal and transverse wind resistance coefficients is shown in Figure 10. It can be seen that in the full-load condition, the coefficient C_{fx} of the container ship with various forecastle fairings is less than that of container ships without a forecastle fairing. Moreover, the coefficient C_{fy} of the container ship with the forecastle fairings is not larger than without a forecastle fairing. This indicated that the forecastle fairings are effective in reducing the wind drag of container ship.

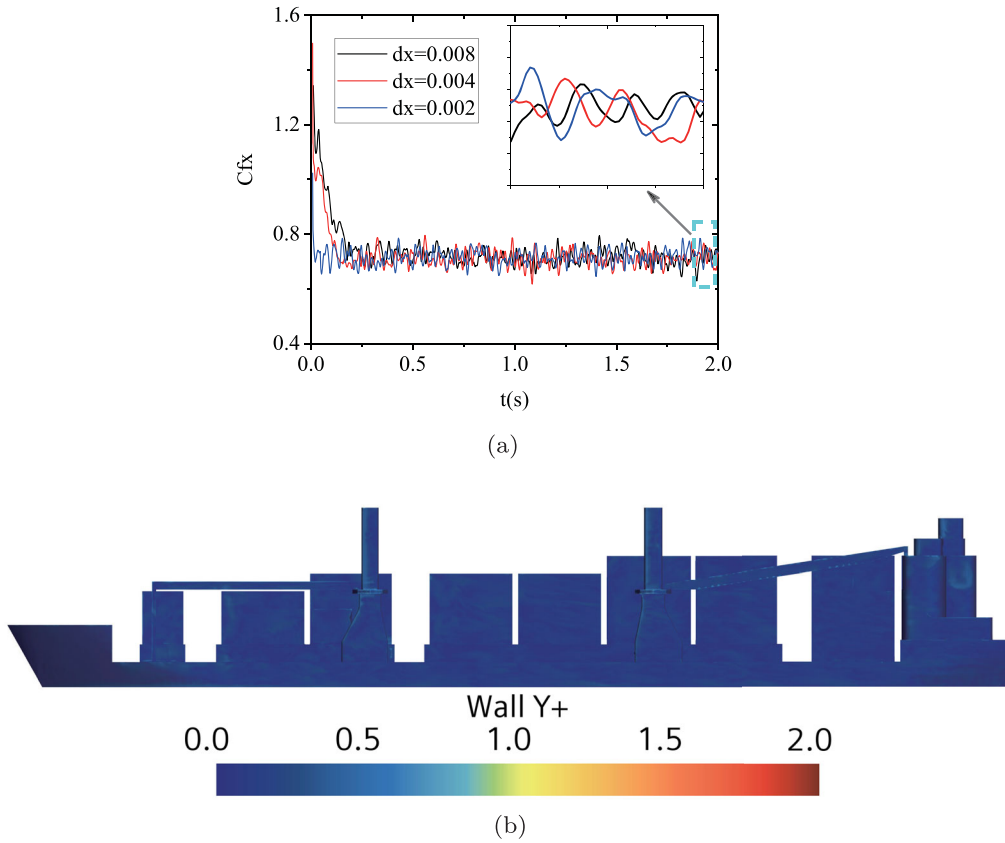


Figure 7. Verification (a) progression of force coefficient through time t for Case A; (b) wall y^+ , $dx = 0.002$.

Table 5. Validation of computational fluid dynamics (CFD) results (coarse, medium, and fine mesh).

Numerical results	Meshes			Experimental data	Error(%)		
	Coarse	Medium	Fine		Coarse	Medium	Fine
C_{fx}	0.729	0.735	0.737	0.758	3.8	3.0	2.8

As container ships generally sail into the wind, the wind resistance for wind direction $\alpha = 0^\circ$ is considerably more important than that for other wind directions. A factor is defined as follows to analyze the drag reduction rate for a container ship:

$$\Delta C_F\% = \frac{C_{Ff} - C_{F0}}{C_{F0}} \times 100\% \quad (18)$$

where C_{F0} is the wind resistance coefficient of the fully loaded ship, and C_{Ff} is the wind resistance coefficient of the container ship with forecastle fairing. The overall drag reduction rates provided by the six forecastle fairings at $U = 25$ m/s ($Re = 4.22 \times 10^6$) are shown in

Table 6. The comparison shows that all six types of forecastle fairings are effective in reducing the wind resistance of the fully loaded container ship model when the wind direction α is 0° or 30° ; in addition, the drag reduction caused by Fairing 2 is more obvious than that of the others, with a drag reduction rate reaching 20.85%. The drag reduction rate gradually decreased as the wind direction α was increased from 0° to 60° , the most pronounced effect being that of Fairing 2 for the change in wind direction from 0° to 30° . When the wind direction reached 60° , the wind resistance could no longer be reduced by any of the six forecastle fairings. These results

Table 6. Drag reduction rate $\Delta C_F\%$.

Wind direction $\alpha (^\circ)$	fairing 1	fairing 2	fairing 3	fairing 4	fairing 5	fairing 6
0	13.66	20.85	13.96	17.78	13.86	11.09
30	6.68	11.21	11.0	11.18	7.49	10.98
60	-26.07	-12.50	-16.59	-15.89	-14.63	-11.96

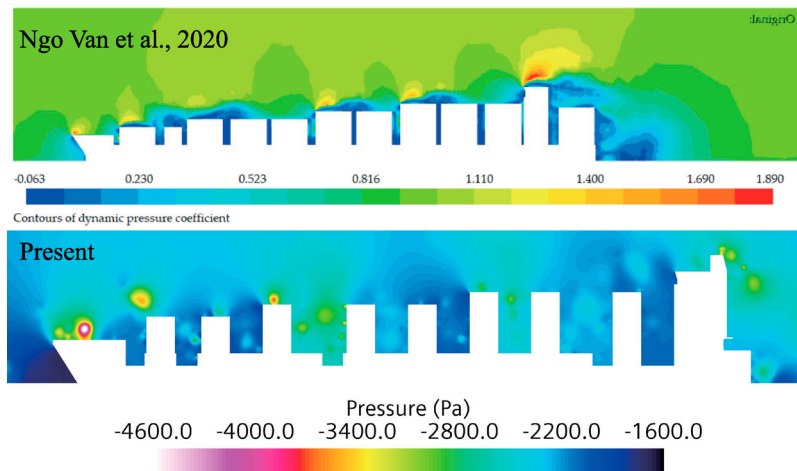


Figure 8. Comparison with the previous studies (Van He et al., 2020) about the pressure distribution.

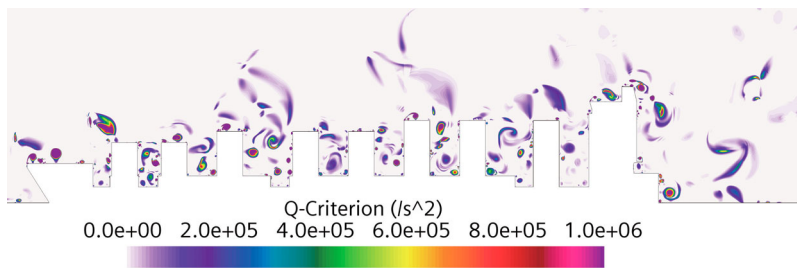


Figure 9. The snapshot of positive and negative vortex at the center section of container ship.

show that the wind resistance of a container ship can be substantially reduced by a well-designed forecastle fairing, and Fairing 2 is the appropriate one for the fully loaded ship.

3.2. Effect of container configuration in the uneven-load condition

The wind resistance coefficients C_{fx} and C_{fy} for the container ship with four types of randomly distributed

loads and a streamlined load (Cases H–L) are shown in Figure 11. the test velocity U is 25 m/s, and the wind direction ranges from $\alpha = 0^\circ$ to $\alpha = 180^\circ$. As shown in the figure, the longitudinal wind resistance coefficient C_{fx} for Case H was greater than those of the others, whereas the transverse wind resistance coefficient C_{fy} was clearly not less than those of the others. This indicates that the configuration in Case H may adversely affect wind resistance. Meanwhile, the lowest longitudinal wind resistance coefficient is that for Case K, and the transverse

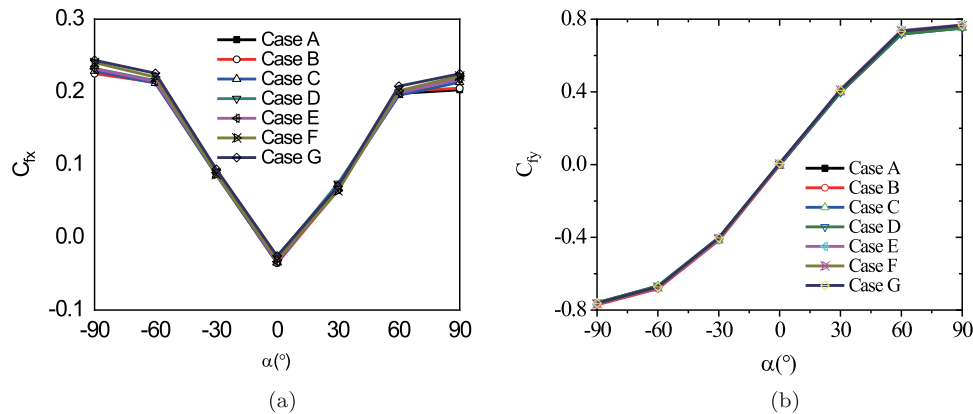


Figure 10. The non-dimensional wind resistances of ship model with different forecastle fairings. (a) longitudinal component (b) transverse component. $U = 25$ m/s, Case A–G. (a) C_{fx} and (b) C_{fy} .

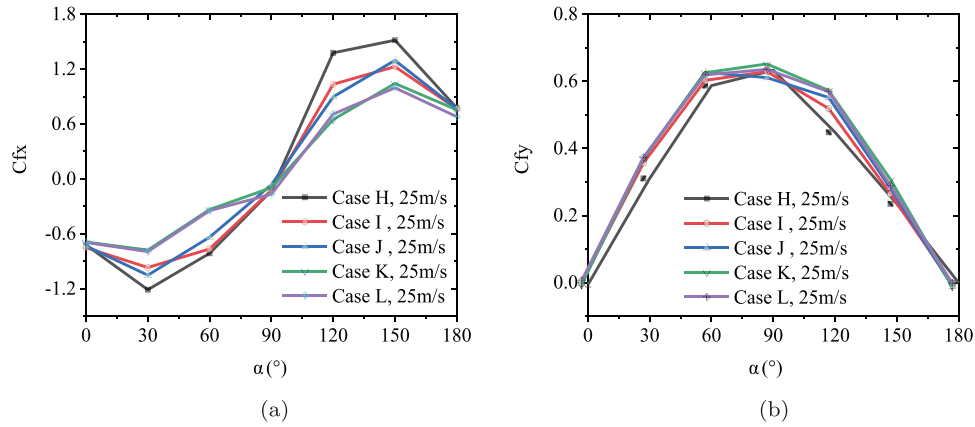


Figure 11. Dimensionless wind resistances of ship model without Fairing 2 for randomly distributed loads (Cases H–K) and streamlined load (Case L): (a) longitudinal component; (b) transverse component. $U = 25$ m/s. (a) C_{fx} and (b) C_{fy} .

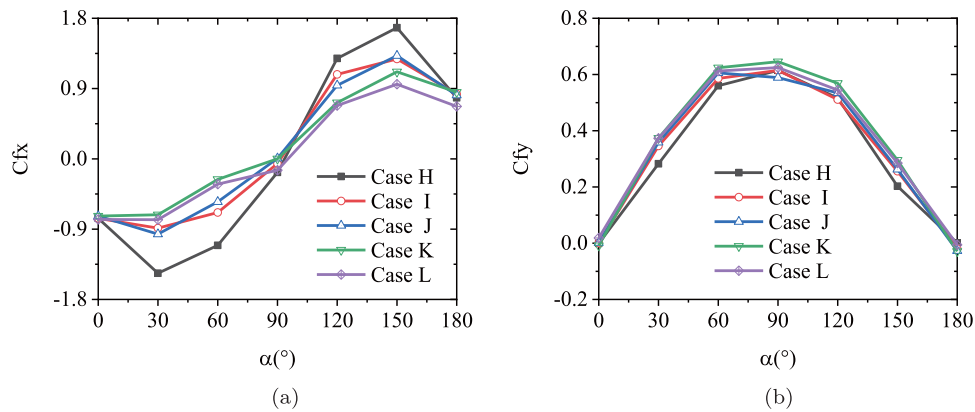


Figure 12. Dimensionless wind resistances of ship model without Fairing 2 for randomly distributed loads (Cases H–K) and streamlined load (Case L): (a) longitudinal component; (b) transverse component. $U = 15$ m/s. (a) C_{fx} and (b) C_{fy} .

wind resistance coefficient for this case is greater than those of the others, although it is not so obvious. The results for Case L, which is for a streamlined load, are similar to those for Case K. The results shown in Figure 8 demonstrate that the wind resistance under uneven loads

can be reduced by certain container configurations and that the streamlined load configuration (Case L) and that of randomly distributed loads in Case K are considerably more effective than the other configurations for this container ship.

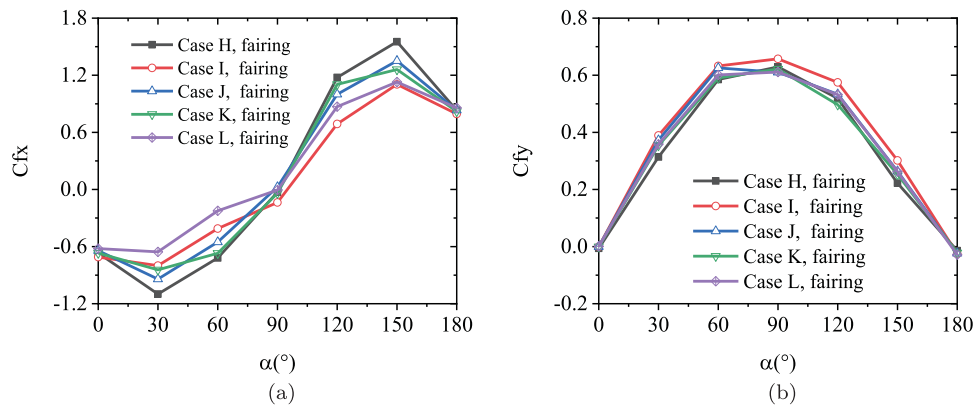


Figure 13. Dimensionless wind resistances of ship model with Fairing 2 for randomly distributed loads (Cases H–K modified to include Fairing 2) and streamlined load (Case L modified to include Fairing 2): (a) longitudinal component; (b) transverse component. $U = 15$ m/s. (a) C_{fx} and (b) C_{fy} .

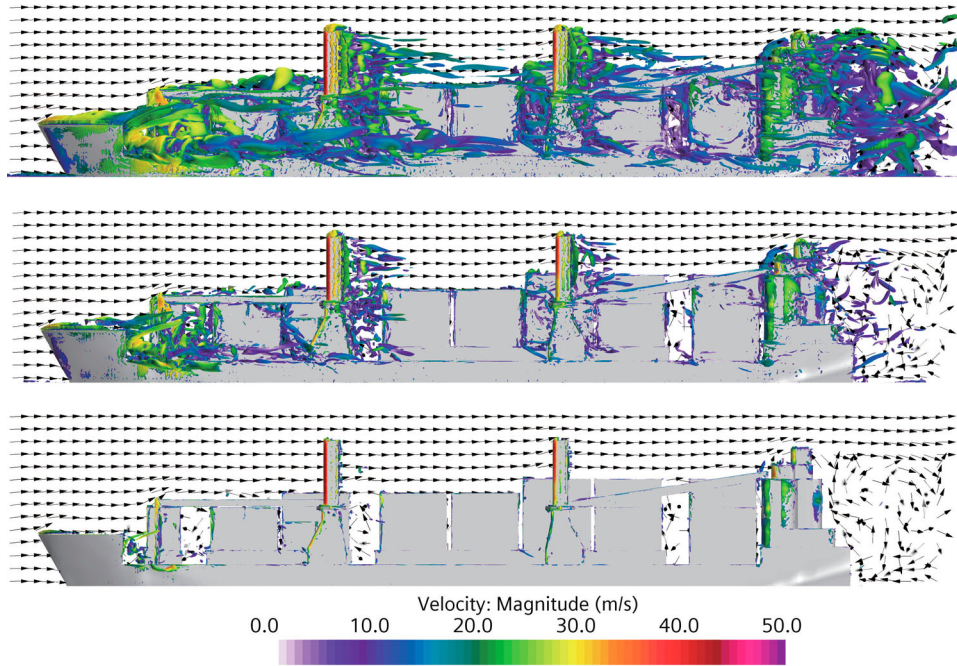


Figure 14. Vortical structure of the fully loaded container ship without forecastle fairing. $Q = 1 \times 10^4 \text{ s}^{-2}$, $Q = 5 \times 10^4 \text{ s}^{-2}$, $Q = 25 \times 10^4 \text{ s}^{-2}$.

3.3. Effect of forecastle fairing in the uneven load condition

The dimensionless wind resistances of the ship model without Fairing 2 for randomly distributed loads and the streamlined load are shown in Figure 12 and the wind velocity U is 15 m/s. It can be seen that at this wind speed, Cases K and L still have the least wind resistance. We know from the findings presented in Section 3.1 that Fairing 2 provides the best drag reduction in the full-load condition; the question arises whether the forecastle fairing is also effective in the uneven-load condition. The dimensionless wind resistances for the ship model in Cases H–L modified to include Fairing 2 are shown in Figure 13.

There is a clear reduction in the wind resistance coefficient from those displayed in Figure 12. In Figure 13, the wind resistance coefficients for Case K and Case L modified to include Fairing 2 are less than those for the other container configurations. Thus, Case K and Case L modified to include Fairing 2 are the better container configurations.

4. Numerical results and discussion

4.1. Clarification of the vortex

The oscillation of the coefficient C_{fx} over time (Figure 7(a)) is caused by the lateral oscillations of the quasi-stable vortices of the flow separation (Hemida & Baker,

2010; Östh & Krajnović, 2014), which are influenced by the forecastle fairing and the containers stacked on the ship. Therefore, the vortex distribution was investigated for the different types of forecastle fairings and container configurations. Vortices are identified by the Q -criterion, which is the second invariant of the velocity gradient tensor in an incompressible flow:

$$Q = \frac{1}{2} \left(\|\Omega\|^2 + \|S\|^2 \right) = \frac{1}{2} \left(\left(\frac{\partial u_i}{\partial x_i} \right)^2 - \frac{\partial u_i}{\partial x_i} \frac{\partial u_j}{\partial y_j} \right) \quad (19)$$

Where Ω and S are the symmetric rotation tensor and antisymmetric rate-of-strain tensor respectively, of the filtered velocity variables. If Q is positive, the vorticity is dominant; otherwise, the strain is dominant. Iso-surfaces of positive values of Q thus reveal locations in the flow where rotation dominates over strain and hence indicate vortical structures in the flow. The vortex field was studied using the iso-surfaces with positive Q values. As air flows around the container ship, impingement of the wind field will occur on the surface formed by the hull and the containers, and a large number of vortices will be formed in three dimensions around the container ship; these are separated from the edges of the hull and the container surfaces. Typical vortical structures are shown in Figure 14, where the Q values of the iso-surface are $Q = 1 \times 10^4 \text{ s}^{-2}$, $5 \times 10^4 \text{ s}^{-2}$, and $25 \times 10^4 \text{ s}^{-2}$, and the iso-surfaces are colored according to the magnitude of the velocity.

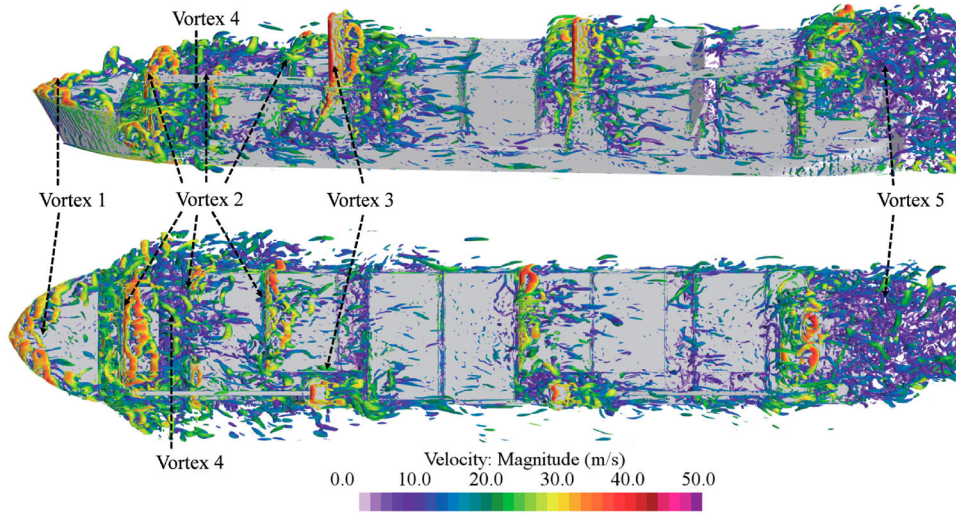


Figure 15. Classification of the vortices. $Q = 5 \times 10^5 \text{ s}^{-2}$.

As Figure 14 shows, the flow field of a three-dimensional container ship is complex. There are dozens of large-scale and small-scale vortices that are generated primarily in the regions at the bow, between the containers, and behind the container ship. The vortices generated at the bow are separated from the lateral sides, those generated between the containers move downstream, and those generated behind the hull rotate in the vertical plane. The complexity of the flow is due to the additional two shear layers emanating from the lateral sides of the body and the greater number of degrees of freedom (Östth & Krajnović, 2014). It is demonstrated that the large-scale vortices are generated at the bow and that most of the small-scale vortices are generated in the other regions. A three-dimensional vortex interaction is observed, and the small-scale complex vortices are dissipated or merge into large-scale vortices.

To clearly and accurately describe the vortex field near the container ship, we selected the value $Q = 5 \times 10^4 \text{ s}^{-2}$ for the vortex. Considering that the vortices are primarily generated in certain typical regions of the flow field, they may be classified into several types as shown in Figure 15. The figure shows side and top views of the vortical structures in the flow field; these are classified by region and the interaction between them. Vortices generated at the bow from the upper deck are called forecastle-deck vortices (vortex type 1, denoted herein as ‘Vortex 1’ vortices), vortices induced by the container blocks with gaps and moving downstream are called container-edge vortices (vortex type 2, denoted herein as ‘Vortex 2’ vortices), and those formed by the blocking effect of the crane are called crane-edge vortices (vortex type 3, denoted herein as ‘Vortex 3’ vortices). The vortices generated upstream and separating merge with these newly generated vortices, forming secondary-flow vortices (vortex type 4,

denoted herein as ‘Vortex 4’ vortices). Eventually, all vortices arrive at the region behind the hull, where they are called trailing vortices (‘Vortex 5’) and rotate in the vertical plane. This vortex classification will be used in studying the influence of forecastle fairing and container configurations on the aerodynamic performance of the container ship.

4.2. Effect of forecastle fairing

4.2.1. Influence on the pressure distribution

It was demonstrated in the model test that Fairing 2 is the forecastle fairing that is most effective in reducing the wind resistance of the fully loaded container ship. In order to investigate the effects of Fairing 2, numerical simulations were performed for Cases A and C. The dimensionless wind pressure C_p is defined as

$$C_p = \frac{P_t - P_\infty}{0.5\rho U_\infty^2} \quad (20)$$

Where P_∞ is the standard atmospheric pressure. U_∞ is the wind velocity. The distributions of pressure on the profiles generated by the intersection of the hull surface and the longitudinal planes at $y = 0$, $y = 0.05$ and $y = 0.1$ m, which are located at the center, $\frac{1}{8}B$ and $\frac{1}{4}B$ in the Y directions (Figure 16), are compared in Figure 17. The figure shows the CFD results for the pressure coefficient distributions along the three lines for Cases A and C at time $t = 10.0$ s. It can be seen that on the whole, the pressure has both positive and negative effects on the surface formed by the hull and container blocks. When the wind passes along the container ship, a variable maximum pressure occurs on the surface of the forecastle fairing. The pressure along the forecastle fairing to the

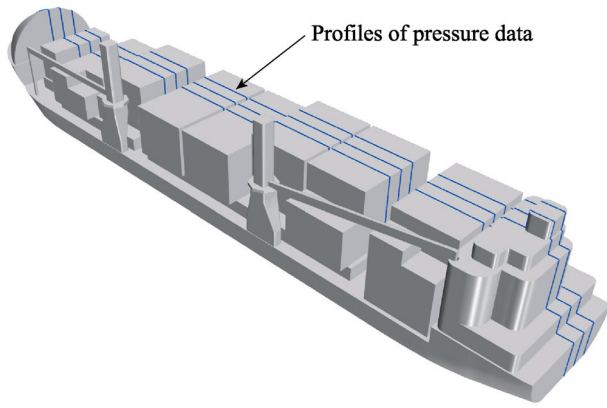


Figure 16. Location of profiles for the plots of pressure coefficient distributions, showing three representative lines around the container ship model.

rear of the ship appears to be small, but variable maximum pressures occur at the edge of the container blocks with gaps. Pressure oscillations at the edge of the container blocks with gaps indicate that Vortex 2 vortices have an effect on the pressure. In contrast to Case A, which has no fairing, Fairing 2 in Case C can effectively reduce the pressure maxima and oscillations at the edge of the container blocks with gaps, although the forecastle fairing has little effect on the pressure distribution at the bow of the ship.

Figure 18 shows the distribution of instantaneous pressures around a set of container blocks at time $t = 10.0$ s. Figure 18(a) shows the location of the container blocks, which are positioned at $0.51L$ in X direction. In Figure 18(b,c), the bottom surface is the projection area for the pressure distribution, and the dotted marking line, which divides the surface of the container blocks into side and upper surfaces, represents the edge of the container blocks. The difference between the maximum and minimum pressures on the surfaces of the blocks is large, reaching 654 Pa and 532 Pa for Case A and Case C, respectively. The maximum pressure is concentrated around the edge of the container blocks. The closer to the edge of the container blocks, the greater the pressure on the container surface.

We also analyzed the mean pressure distribution around the container blocks with gaps from time $t = 9.0$ s to $t = 11.0$ s for Cases A and C, with the results as shown in Figure 19. In the figure, the blue region indicates low pressure, and the red region indicates high pressure (red and yellow colors). The high-pressure area in Figure 19(b) has been markedly reduced from the mean pressure displayed in Figure 19(a). Figures 18 and 19 demonstrate that Fairing 2 has an important impact on the distribution of pressure on container blocks with gaps.

4.2.2. Influence on the flow fields

To more fully understand the effect of the forecastle fairings, details of the flow field for Cases A and C were analyzed. The instantaneous streamlines around the container ship as viewed from the side were obtained by numerical simulation; these are shown in Figure 20. The wind velocity was 25 m/s, and the wind direction was 0° . The surfaces of the container ship are colored according to pressure. The streamlines are higher at the bow with Fairing 2 (Case C) than without (Case A). Thus, Fairing 2 has an effect in guiding the wind. The velocity field shown in Figure 20(Case C) around the container blocks with gaps is also substantially weakened, particularly at the bow of the ship. It can be seen that the pressures distributed on the surfaces of the ship in Figure 20(Case C) are less than those in Figure 20(Case A).

Figure 21 presents the instantaneous vortical structures around the container ship as viewed from the side, for Cases A and C. The iso-surfaces of the Q -criterion are colored by the velocity magnitude and the value of Q -criterion are $Q = 5 \times 10^5 \text{ s}^{-2}$. It can be seen that the wind is blocked by the surface formed by the ship and containers, and flow separation occurs at the bow, forming Vortex 1 vortices of different sizes. When there is no forecastle fairing (Figure 21(a)), the vortical structures on the surface of the ship and containers are quite obvious, and the flow separation is more perceptible at the edge of the container blocks with gaps, forming Vortex 2 vortices. In addition, when the wind passes along the container ship, there are vortex interactions around the surface of the ship and containers, which cause many small-scale Vortex 4 vortices to be generated, and the presence of a large number of vortices is clearly not conducive to reducing the drag of the container ship. The vortical structure in Figure 21 explains the pressure distributions with and without forecastle fairing analyzed in Section 4.3.1. The number of vortices was markedly reduced after the container ship (Case A) was equipped with Fairing 2 (Case C). Consequently, it is concluded that the use of Fairing 2 is a highly effective way to reduce the wind resistance of a container ship.

4.3. Effect of container configuration

4.3.1. Influence on the pressure distribution

As demonstrated by the results of the wind tunnel experiment (Section 3), the wind resistance is the highest for Case H, and Cases K and L exhibit better drag reduction. The distributions of pressure around container ships with different container configurations (Cases H, K, and L) are shown in Figure 22; the representative locations for the pressure data obtained by numerical simulation are similar to those given in Figure 16. As shown in Figure 22,

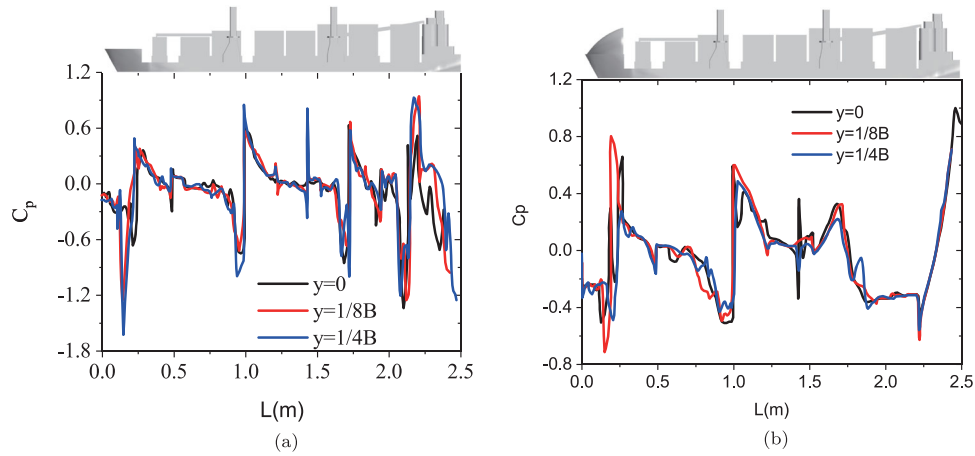


Figure 17. Comparison of the distribution of pressure coefficient C_p on the three selected profiles. $U = 25$ m/s. (a) Case A and (b) Case C.

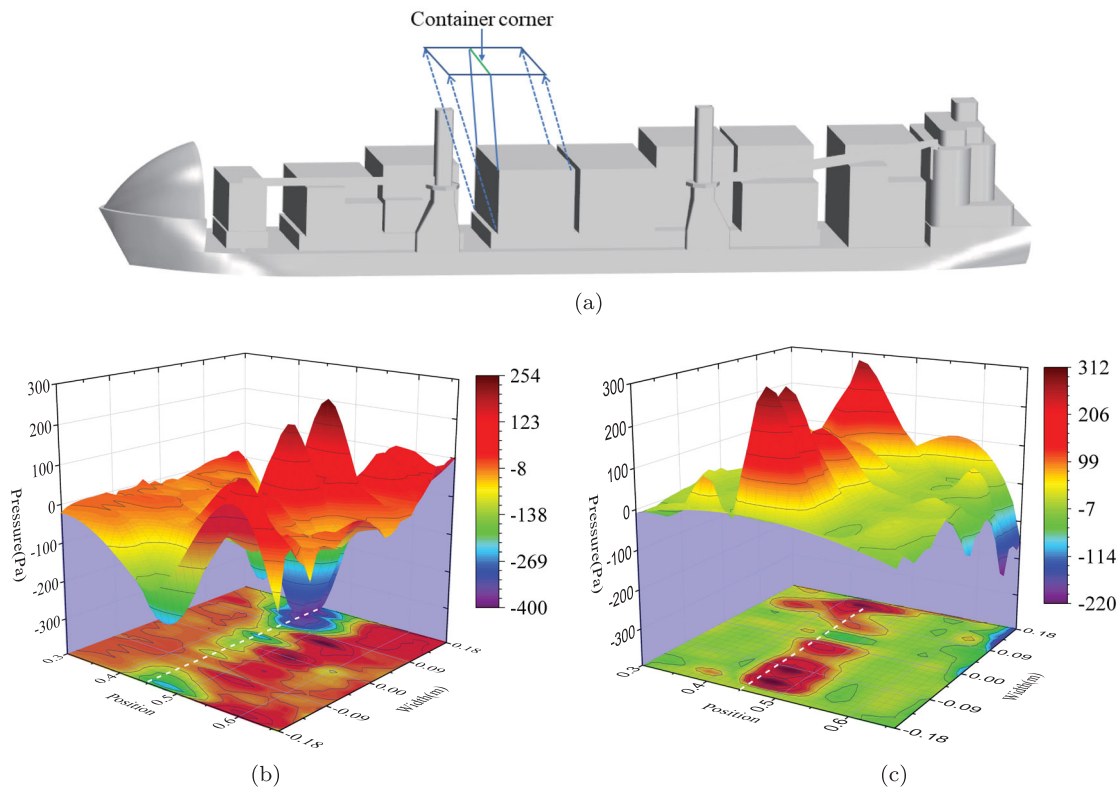


Figure 18. Distribution of instantaneous pressure around a set of container blocks. The dotted line in (b) and (c) indicates the edge between side and upper surfaces. $U = 25$ m/s. (a) Location of container blocks. (b) Without forecastle fairing (Case A) and (c) With Fairing 2 (Case C).

there are some pressure oscillations at the edge of the container blocks with gaps. The oscillations are considerably weaker for Cases K and L than for Case H, although the variable maximum pressure at the edge of the container blocks with gaps does not decrease. In addition, the pressure for Case K was reduced to near zero at the rear of the hull, where there are no containers. It can be seen that in contrast to Case H, Case L has negative pressure at the rear of the hull.

We also studied the distribution of instantaneous pressure for Cases H, K, and L in a representative location similar to that given in Figure 18(a), with the results as shown in Figure 23. In contrast to the results for the randomly distributed load in Case H, the number of pressure maxima in Cases K and L was dramatically reduced at the edge of the container blocks. Such effects by the container configurations will change the wind drag acting on the container ships. The configurations

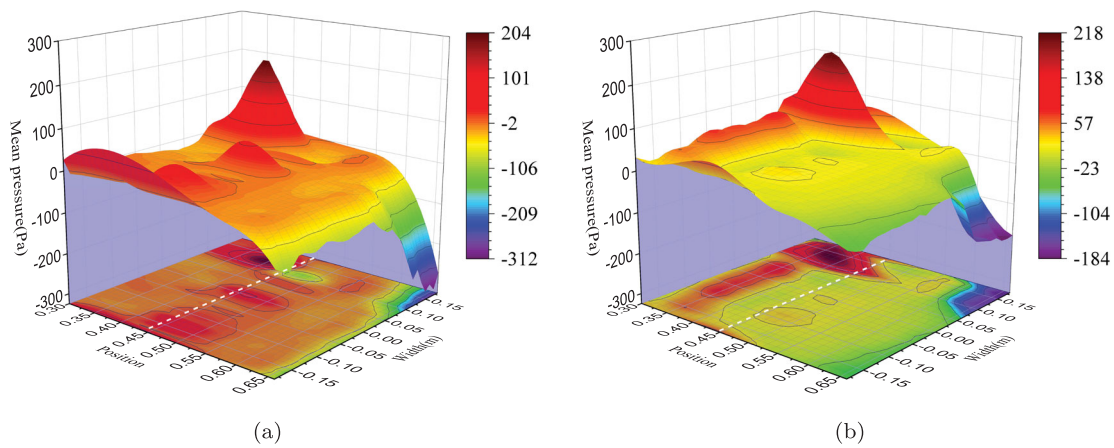


Figure 19. Mean pressure distribution around the container blocks. $U = 25$ m/s. (a) Without forecastle fairing (Case A) and (b) With Fairing 2 (Case C).

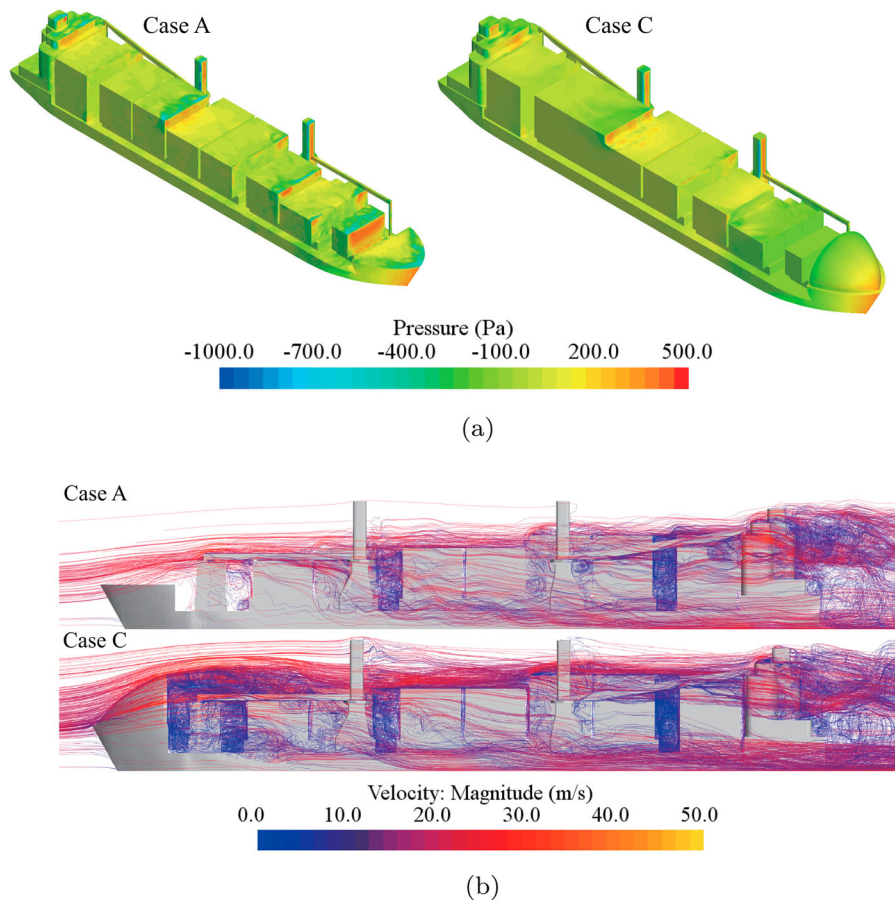


Figure 20. Comparison of the distribution of pressure on the surface of container ship and instantaneous streamlines around the container ship for Case A and Case C. $U = 25$ m/s. (a) Distribution of pressure and (b) The instantaneous streamlines.

of the streamlined load and the randomly distributed load with no containers aft are conducive to reducing the pressure on the surface of container blocks with gaps.

In addition, the mean pressure distribution around the container blocks with gaps is shown in Figure 24 for the container configurations in Cases H, K, and L. As can

be seen, when the wind passes along the container ship, the high-pressure region around the container blocks that appears on the surface of the blocks increases the wind drag, a finding consistent with the numerical results shown in Figure 19. Among the three cases compared here, the pressure in Case L is substantially less than that in Case H or Case K.

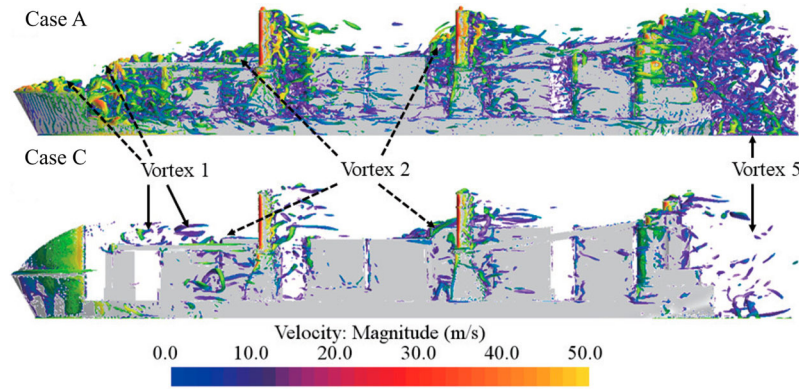


Figure 21. Instantaneous vortical structures around the container ship (side view) for Case A and Case C. The iso-surfaces of the Q -criterion are colored according to the velocity magnitude. $Q = 5 \times 10^5 \text{ s}^{-2}$.

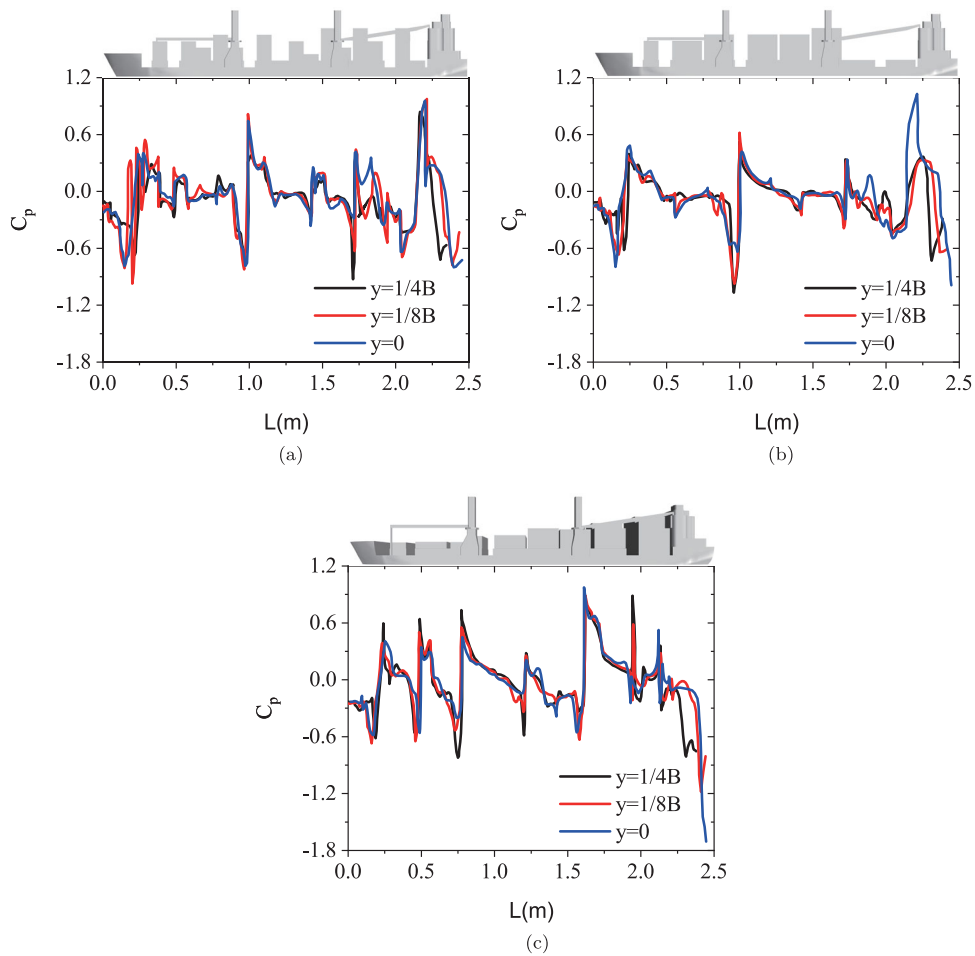


Figure 22. Comparison of the distribution of the pressure coefficient values on the three selected container ship profiles. $U = 25 \text{ m/s}$. (a) Case H. (b) Case K and (c) Case L.

4.3.2. Influence on the flow fields

The distribution of pressure and instantaneous streamlines around the container ship as viewed from the side are shown in Figure 25. The wind velocity is 25 m/s , and the wind direction was 0° . The surfaces of the container

ships are colored according to pressure. In Case K (Figure 25), the containers are concentrated in the front and middle parts of the ship, resulting in a relatively low surface area and gaps between the container blocks. Therefore, the wind resistance decreased at the end of the

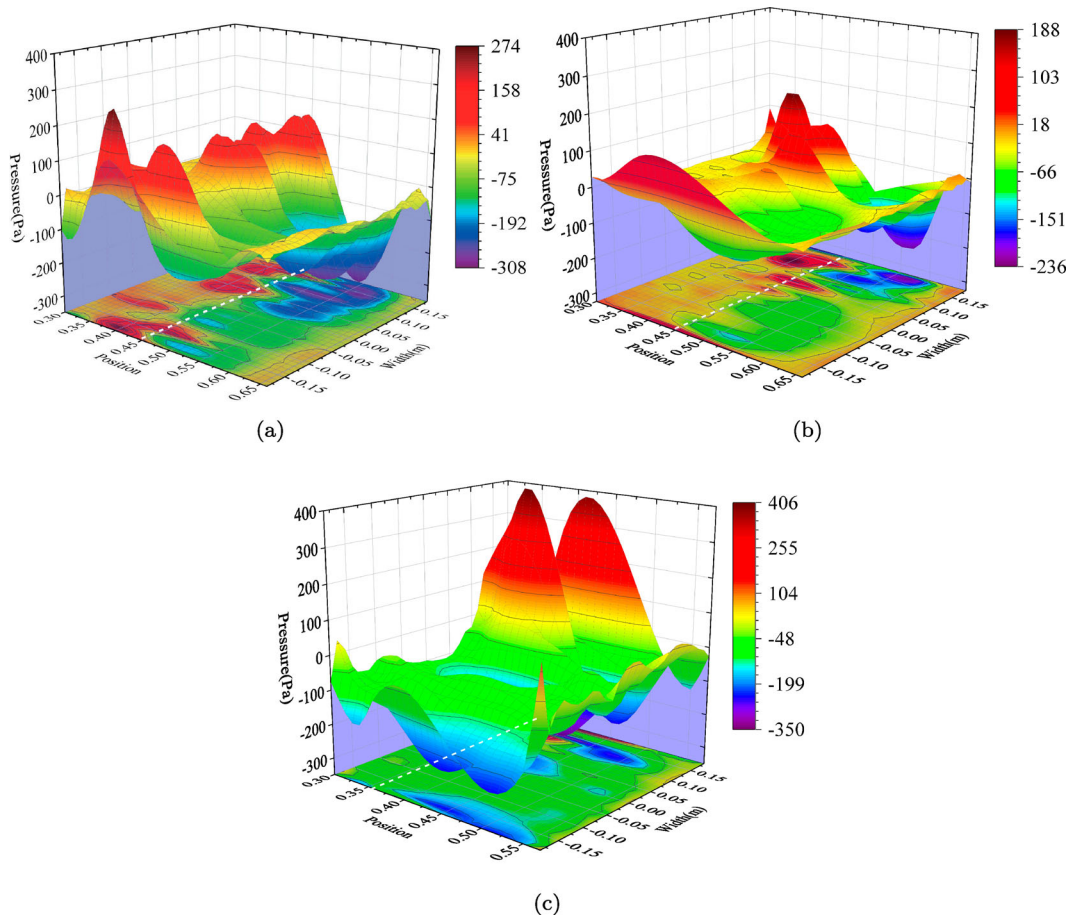


Figure 23. Distribution of instantaneous pressure around the container blocks. $U = 25$ m/s. (a) Case H. (b) Case K and (c) Case L.

container ship as a result of the influence of the container blocks, although the effect is only a slight change in the streamlines and pressure in the front and middle parts of the ship, in contrast to the effect in Case H, with no containers aft. Furthermore, the streamlines in Cases H and K are more complex than those in Case L (Figure 25) owing to the large windward surface of the container blocks. In Case L, the stack height of the containers increases gradually and regularly, giving Case L a substantial advantage in propagating the streamlines around the ship.

The vortical structures for Cases H, K, and L are presented in Figure 26. The iso-surfaces of the Q -criterion are colored according to the velocity magnitude; here, the Q -criterion Q is $Q = 5 \times 10^6 \text{ s}^{-2}$. In general, the forces on the ship are expected to increase with an increase in the surface area of container blocks with gaps. As shown in the figure, the container configurations in Cases K and L both reduced the blocked surface area on the container ship, where the flow impinges directly on the ship surface. Although the vortical fields in the front and middle parts of the ship in Case K differ only little from those in Case

H, large amounts of vortex interaction have been avoided, and thus the flow separation at the surface of the containers and ship is avoided as well. In addition, the vortical fields in Case L from bow to stern are clearly smaller than those in Case H.

Massive Vortex 1 vortices induced by the hull surface are generated at the bow of the container ship, and the three container configurations have a major impact on Vortex 2, which is induced by the container blocks with gaps, as shown in Figure 26(Case H). The drag reduction for the container ship in Case K is due primarily to the concentration of container blocks, which reduces the impact of a series of contiguous container gaps, resulting in a reduction in the number of Vortex 2 vortices (Figure 26(Case K)). Therefore, this is the main reason for the drag reduction in Case K. As the change in container stack heights is relatively smooth, the containers have less impact on the vortical structures. The vortex in Figure 26(Case L) propagates easily downward from the bow to the stern of the ship. The Vortex 2 vortices are reduced in number, and the vortex interaction is substantially weakened; this is the main reason for the drag reduction

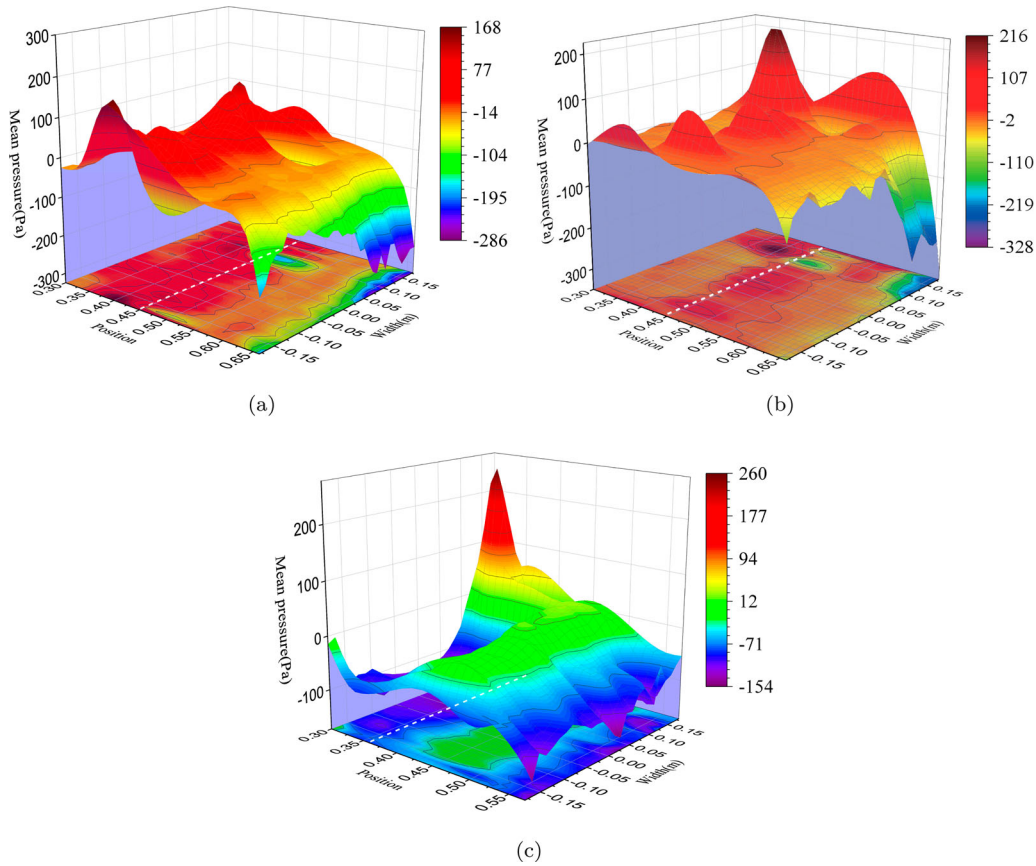


Figure 24. Mean pressure distribution around the container blocks. $U = 25$ m/s. (a) Case H. (b) Case K and (c) Case L.

in Case L. In summary, for scenarios in which a container ship cannot be guaranteed to remain fully loaded, the numerical simulation explains why the randomly distributed load with no containers aft and the streamlined load offer less wind resistance than the other container ship configurations investigated in the wind tunnel experiments.

4.4. Joint effect of container configuration and forecastle fairing under uneven load

4.4.1. Influence on the pressure distribution

Lastly, we investigated the effect of the different container configurations with and without a forecastle fairing and obtained the numerical simulation results for Case K, Case L, Case K modified to include Fairing 2, and Case L modified to include Fairing 2. Figure 27 shows the pressure distribution around the container ship with and without Fairing 2 at the representative locations at the center, $\frac{1}{8}B$ and $\frac{1}{4}B$. As shown in Figures 27(a–d), the presence of the fairing makes little difference in the distribution of pressure around the container ship except for the location of the maximum pressure, which appears mainly at the edge of the container blocks with gaps. The

main effect of Fairing 2 is that the value of the pressure at the stern is changed from negative to positive, indicating that Fairing 2 changes the pressure distribution for the container ship. Overall, when a container ship is not fully loaded, the distribution of pressure on the ship is affected more by the container configuration than by the forecastle fairing.

Next, we conducted a detailed analysis of the pressure distribution on the surfaces of the container blocks. Figure 28 shows the instantaneous pressures for Case K, Case L, Case K modified to include Fairing 2, and Case L modified to include Fairing 2. The overall pressure for Case K modified to include Fairing 2 (Figure 28(c)) is slightly less than that for Case K (Figure 28(a)), indicating that the forecastle fairing has the effect of reducing the pressure at the gaps between the container blocks in the container configuration of a randomly distributed load with no containers aft. However, by comparing Figure 28(b,d), it can be seen that the forecastle fairing has no obvious influence on the pressure in the streamlined container configuration. Overall, both the forecastle fairing and the container configuration will affect the dynamic pressure on the surface of the container blocks with gaps.

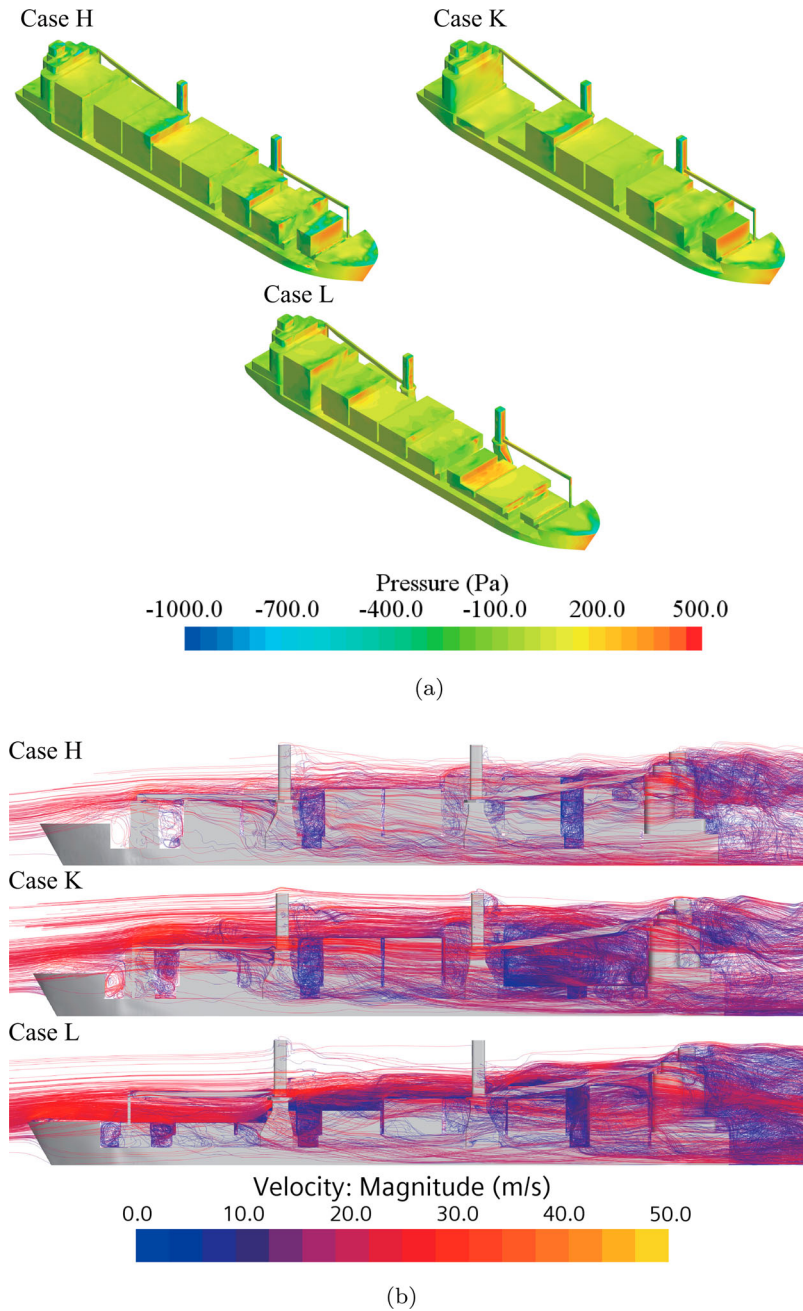


Figure 25. Comparison of the distribution of pressure on the surface of container ship and the instantaneous streamlines around the container ship for Case H, Case K and Case L. $U = 25$ m/s. (a) Distribution of pressure and (b) The instantaneous streamlines.

Figure 29 shows the mean pressure distribution around the container blocks with gaps for Case K, Case L, Case K modified to include Fairing 2, and Case L modified to include Fairing 2. A comparison of Figure 29(a,b) shows that the overall mean pressure for Case L at the position of the monitored container blocks was lower than that for Case K. Furthermore, by comparing Figure 29(c,d), we can see that after being equipped with Fairing 2, the mean pressure for the Case L configuration was lower than that for the Case K configuration. This shows

that the streamlined load is more conducive to reducing the pressure at gaps between the container blocks.

4.4.2. The influence on the flow fields

We performed the numerical simulation analysis on the pressure, streamlines and vortex fields of the container ship for Case K, Case L, Case K modified to include Fairing 2, and Case L modified to include Fairing 2. The distribution of pressure and the instantaneous streamlines are shown in Figure 30. Compared with Case K and

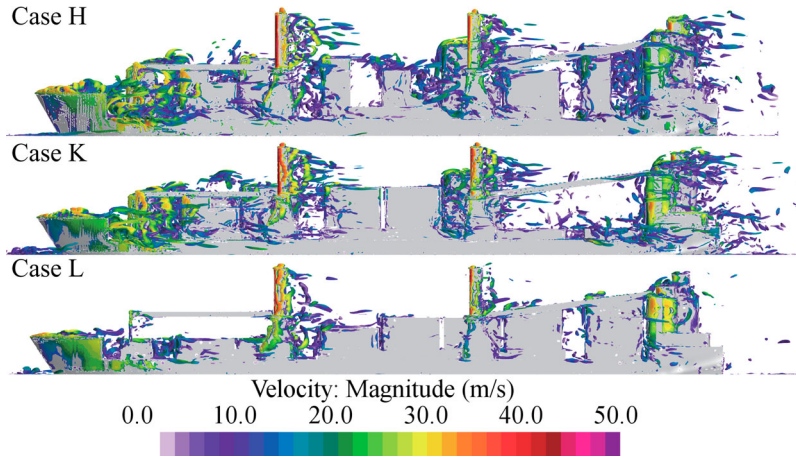


Figure 26. Instantaneous vortical structures around the container ship (side view) for Case H (a randomly distributed load), Case K (a randomly distributed load with no containers aft), and Case L (the streamlined load). The iso-surfaces of the Q -criterion are colored according to the velocity magnitude. $Q = 5 \times 10^6 \text{ s}^{-2}$.

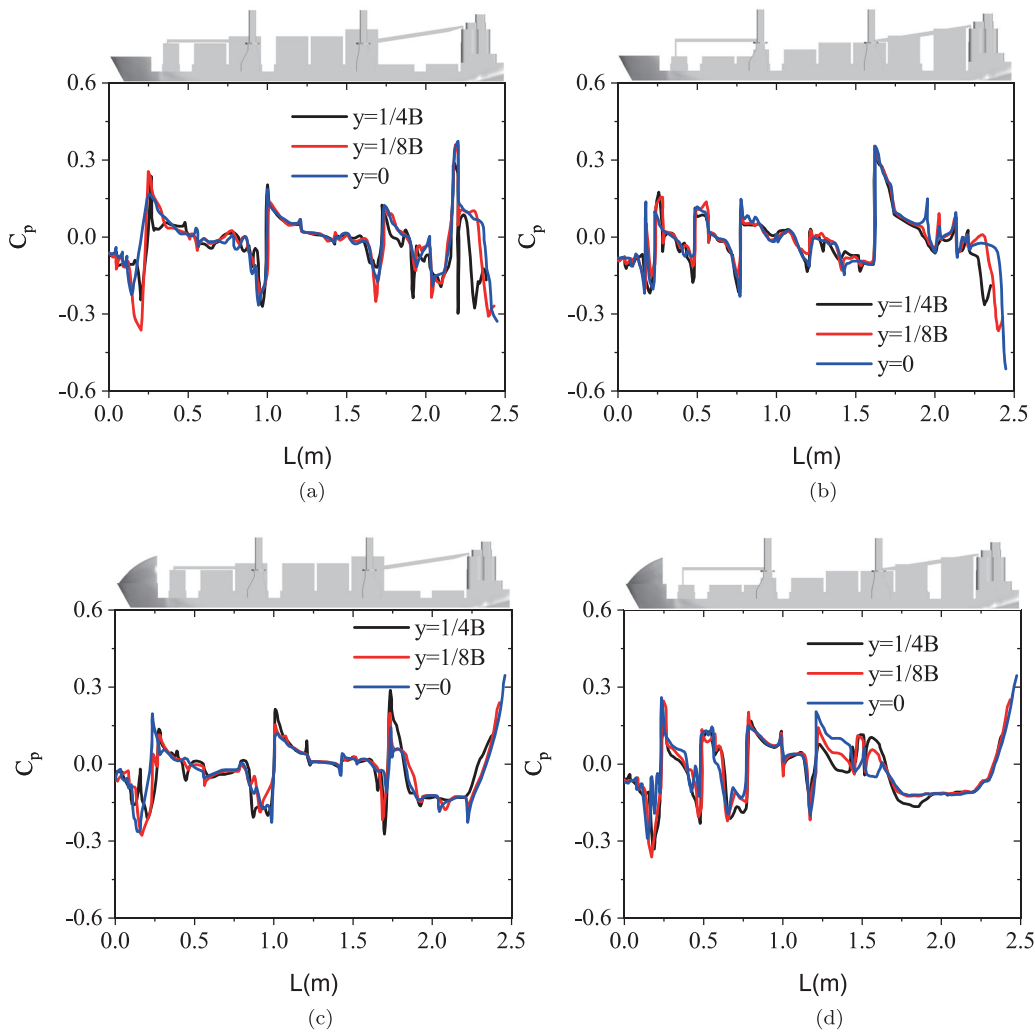


Figure 27. Comparison of the distribution of the pressure coefficient values on the three selected container ship profiles. $U = 15 \text{ m/s}$. (a) Case K (b) Case L, 15 m/s. (c) Case K modified to include Fairing 2 and (d) Case L modified to include Fairing 2.

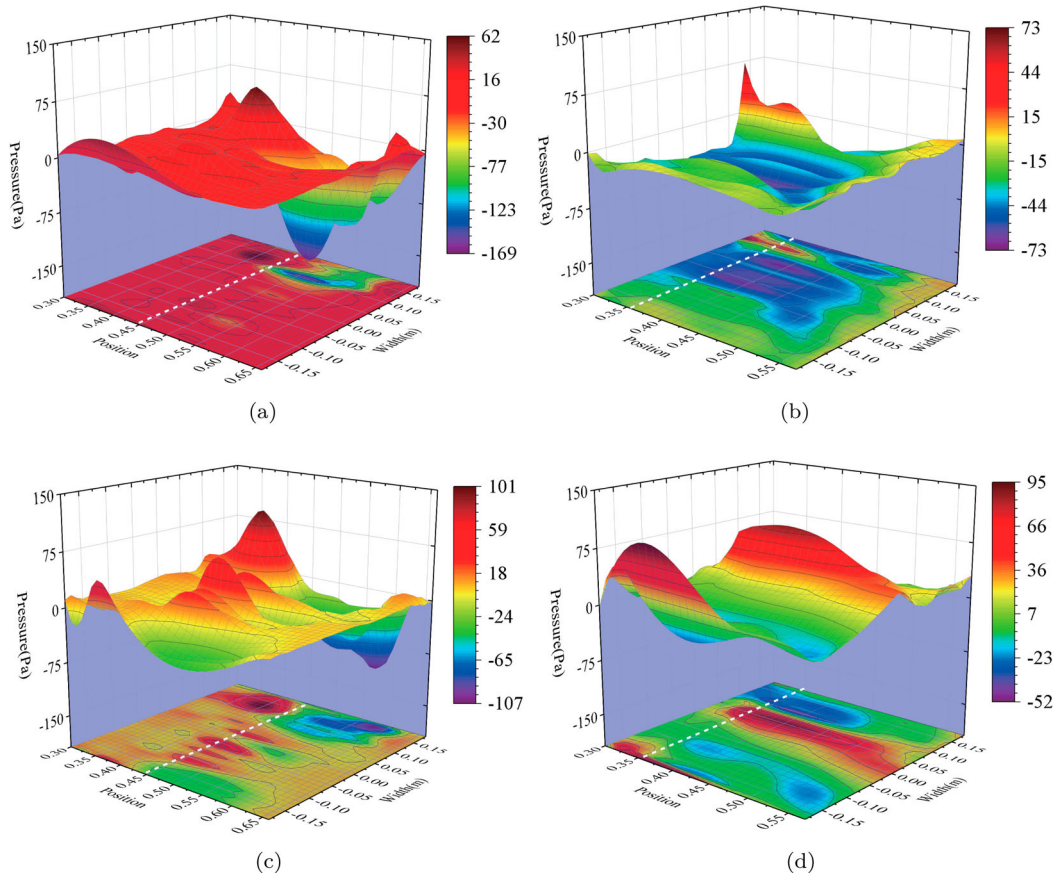


Figure 28. Distribution of instantaneous pressure around the container blocks. $U = 15$ m/s. (a) Case K. (b) Case L. (c) Case K modified to include Fairing 2 and (d) Case L modified to include Fairing 2.

Case K modified to include Fairing 2, the high-pressure area on container ship can be apparently reduced after the Fairing 2 is installed. The streamlines results seen in Figure 30(Case L) were clearly hindered by the container blocks with gaps at the bow and midship, but the influence of the rear of the ship was weakened. The streamlines in Figure 30(Case L) at the stern of the container ship were affected. It can be seen from Case K modified to include Fairing 2 and Case L modified to include Fairing 2, showing the results with the forecastle fairing included, that Fairing 2 affects the guiding streamlines, reducing their hindrance by the container blocks with gaps. The results displayed in Case K and Case K modified to include Fairing 2 show that the container ship in Case K modified to include Fairing 2 was effective in reducing the hindering effect of container blocks on the streamlines near the bow. Similarly, by comparing Case L and Case L modified to include Fairing 2, we can see that the streamlines in Case L modified to include Fairing 2 were likewise less hindered by the container blocks.

Figure 31 shows the instantaneous vortical structures around the container ship for the same four cases (Case K, Case L, Case K modified to include Fairing 2, and Case L modified to include Fairing 2). Vortex 1 in Figure

31(Case K) is especially obvious at the bow of the ship. However, the main vortex in Figure 31(Case L) is in the gap of the container ship, and Vortex 2 is relatively diffuse. After being equipped with the forecastle fairing, the influence on the vortex was weakened, but this is not obvious by comparing Figure 31(Case K) and (Case K modified to include Fairing 2). The main difference caused by introducing the forecastle fairing was the increase in the height of Vortex 1 at the bow of the ship; the contact between Vortex 1 and Vortex 2 vortices on the surface of the container ship was also alleviated, and the vortex interaction was apparently weakened on the surface of the container ship. In summary, the container configuration and the forecastle fairing both have a decisive influence on the vortex.

5. Conclusions

In this study, the wind resistance characteristics of large container ships with different forecastle fairings and container configurations were evaluated using wind tunnel tests and numerical investigations, the numerical method was verified against the experimental data, and the effects

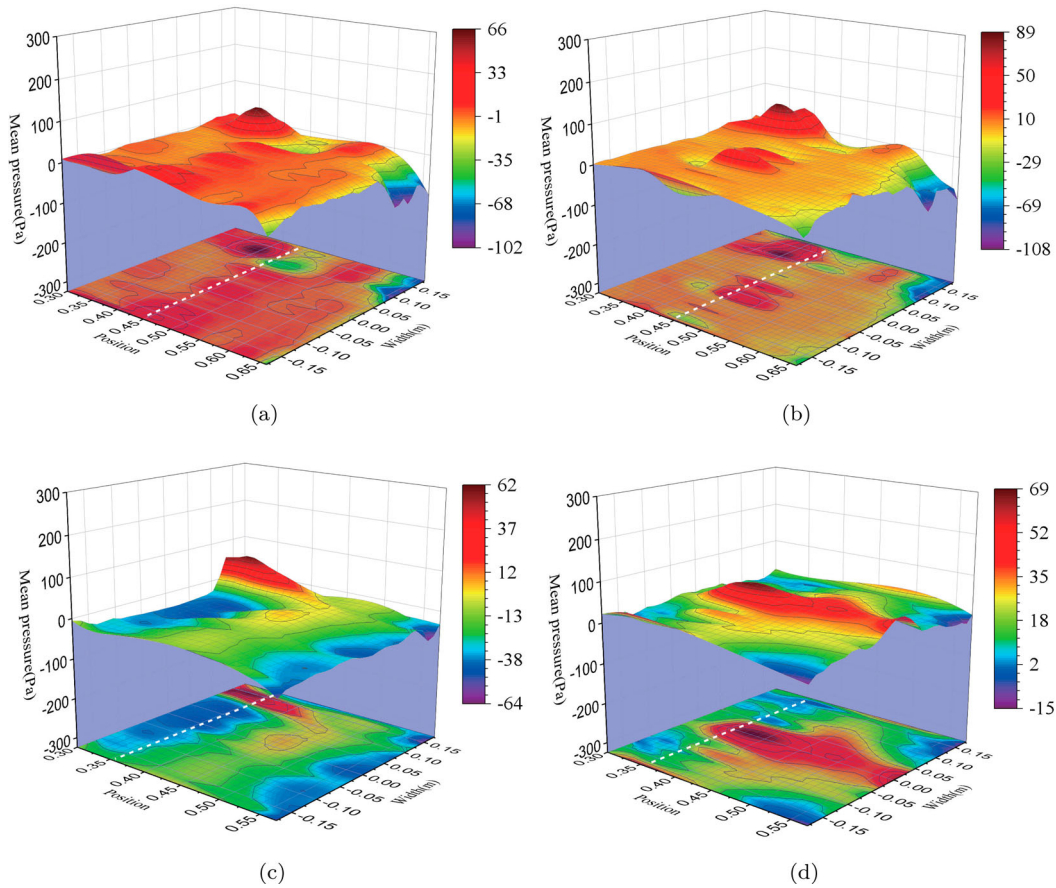


Figure 29. Mean pressure around the container blocks. $U = 15$ m/s. (a) Case K. (b) Case L. (c) Case K modified to include Fairing 2 and (d) Case L modified to include Fairing 2.

on vortical structures and air flow were investigated. The main findings of the study can be summarized as follows.

The forecastle fairings can substantially reduce container ship drag. Among the six forecastle fairing designs studied, Fairing 2 provides the best drag reduction reaching 20.85%. The forecastle fairing and container configuration have a decisive influence on the vortices. The main effect of the forecastle fairing was to increase the height of Vortex 1 at the bow of the ship; it also alleviated the contact between Vortex 1 and Vortex 2 vortices on the ship surface and weakened the vortex interaction at the ship surface.

As the vortices cause the maximum pressure to occur at the edges of container blocks, the presence of container blocks with gaps is not conducive to reducing the drag of the ship. With a full load, the high-pressure area around the container blocks is smaller when the ship is equipped with a forecastle fairing, and with Fairing 2 installed at the bow, the vortices are substantially weakened and fewer in number.

With uneven loads, the wind resistance depends on the stacks of containers. Of the six container configurations studied, the drag coefficients for the streamlined

load and the randomly distributed load with no containers aft are the smallest. For scenarios in which a container ship cannot be guaranteed to maintain a full load, a streamlined load or a randomly distributed load with no containers aft is conducive to reducing the pressure on the surface of container blocks with gaps. The numerical simulation explains why these configurations offer less wind resistance (as demonstrated in the wind tunnel experiments): The randomly distributed load with no containers aft is characterized mainly by the absence of containers at the stern, which reduces the impact on Vortex 2; the streamlined load presents a smooth transition of container heights, which reduces the interaction between Vortex 1 and Vortex 2 vortices.

We recommend that the container ship leave the port with the full load as much as possible. For scenarios in which a container ship cannot be guaranteed to maintain a full load, a streamlined load or a randomly distributed load with no containers aft is suggest to load. In addition, when unloading or reloading halfway at the port, we should keep these configurations. Therefore, it is necessary to divide the packing position reasonably according to the freight task in advance before reloading.

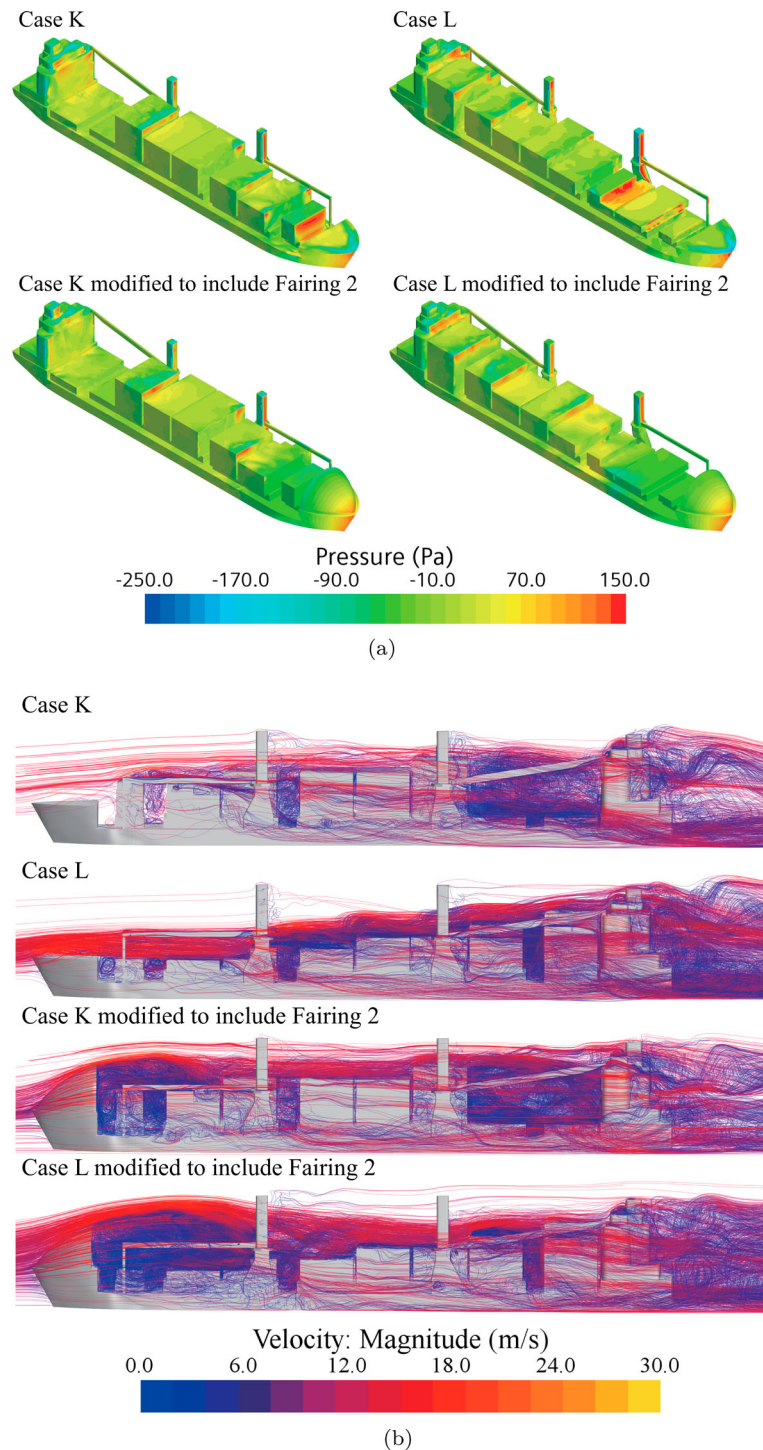


Figure 30. Comparison of the distribution of pressure on the surface of container ship and the instantaneous streamlines around the container ship for Case K, Case L, Case K modified to include Fairing 2 and Case L modified to include Fairing 2. $U = 15$ m/s. (a) Distribution of pressure and (b) The instantaneous streamlines.

6. Inadequate and prospects

Only drag coefficients were included in the validation of the numerical method in the paper. In future investigations, the interaction of the various vortical structures in the gaps between container blocks should be included in PIV experimental test to verify the flow field; The

Reynolds number reached $Re = 4.22 \times 10^6$ when the velocity was 25 m/s in the paper. The airflow in our wind tunnel experiment was a fully developed turbulence flow. However, the wind velocity may be less than 25 m/s and the air flow is not the fully developed turbulence flow near the ground, the impact of wind profiles

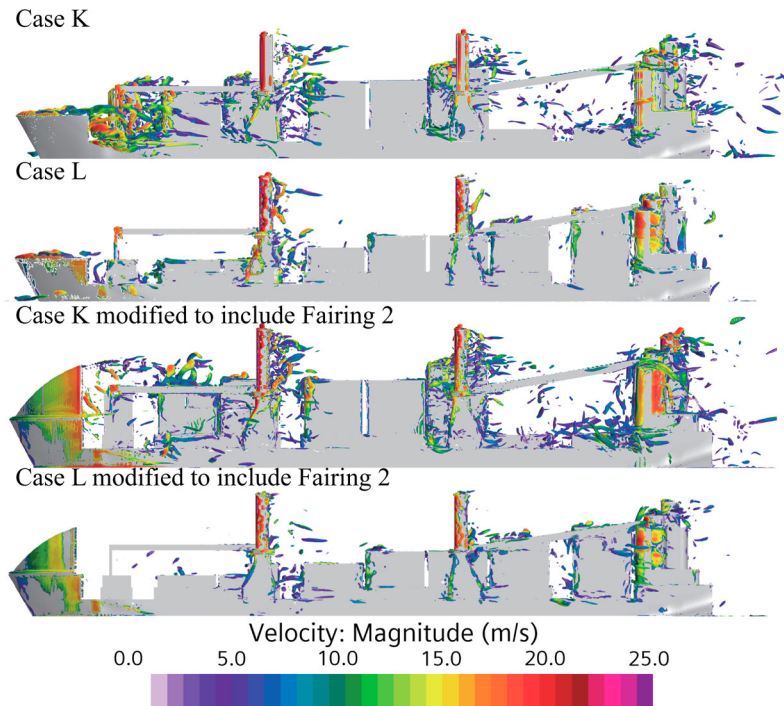


Figure 31. Instantaneous vortical structures around the container ship (side view) for Case K, Case L, Case K modified to include Fairing 2 and Case L modified to include Fairing 2. The iso-surfaces of the Q -criterion are colored according to the velocity magnitude. $Q = 5 \times 10^5 \text{ s}^{-2}$.

and wind directions on vortical structures with various wind velocity in the next work. In the sea, the wind suffered by the superstructure of the container ship is not the uniform boundary condition. Considering the pulsating wind field, the conditions of different inlet turbulence characteristics and other non-uniform flow conditions, the wind resistance and airflow field characteristics of the container ship will be discussed and studied. When the container ship sails on the sea with waves, the ship will produce significant pitch and heave motion. Considering the influence of the ship's pitch and heave degree of freedom motion on the wind resistance coefficient and airflow field, it will be carried out in the future work.

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